

Global Smoothness for the Three-Dimensional Incompressible Navier–Stokes Equations

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Theorem. *For every $\nu > 0$ and every $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, there exists a pressure-normalized pair*

$$(u, p) \in C^\infty([0, \infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R})$$

satisfying

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0, \quad \lim_{|x| \rightarrow \infty} p(t, x) = 0,$$

and

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 \, ds = \frac{1}{2} \|u_0\|_{L^2}^2 \quad (t \geq 0).$$

This pair is unique among pressure-normalized solutions whose velocity belongs to $C([0, T]; H_\sigma^3(\mathbb{R}^3))$ for every finite $T > 0$.

1 Analytic framework

Definition 1.1. For $f \in \mathcal{S}(\mathbb{R}^3)$, set

$$\widehat{f}(\xi) := (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix \cdot \xi} f(x) \, dx, \quad \langle \xi \rangle := (1 + |\xi|^2)^{1/2}.$$

The same integral defines $\widehat{f}$ for $f \in L^1(\mathbb{R}^3)$. The Fourier transform on $\mathcal{S}'(\mathbb{R}^3)$ is its extension by duality. For $s \in \mathbb{R}$, define

$$H^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \langle \xi \rangle^s \widehat{f} \in L^2(\mathbb{R}^3) \right\},$$

and define the finite homogeneous-norm domain by

$$\dot{H}^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \begin{array}{l} \widehat{f} \text{ is represented by a measurable function on } \mathbb{R}^3, \\ |\xi|^s \widehat{f} \in L^2(\mathbb{R}^3) \end{array} \right\}.$$

On these domains set

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} \left| \widehat{f}(\xi) \right|^2 \, d\xi, \quad \|f\|_{\dot{H}^s}^2 := \int_{\mathbb{R}^3} |\xi|^{2s} \left| \widehat{f}(\xi) \right|^2 \, d\xi.$$

Set $A := (1 - \Delta)^{1/2}$, $\Lambda := (-\Delta)^{1/2}$,

$$H_\sigma^s := \{v \in H^s(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}, \quad H^\infty := \bigcap_{m \in \mathbb{N}_0} H^m.$$

$$\mathcal{S}_\sigma(\mathbb{R}^3) := \{v \in \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

For a vector or tensor $F = (F_I)_I$, all Sobolev norms are componentwise:

$$\|F\|_{H^s}^2 := \sum_I \|F_I\|_{H^s}^2, \quad \|F\|_{\dot{H}^s}^2 := \sum_I \|F_I\|_{\dot{H}^s}^2.$$

Repeated Latin indices are summed from 1 to 3.

Lemma 1.2 (Sobolev nesting and embedding). *If $s_2 \geq s_1$, then*

$$\|f\|_{H^{s_1}} \leq \|f\|_{H^{s_2}}.$$

If $k \in \mathbb{N}_0$ and $s > k + 3/2$, then

$$\max_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^\infty} \leq C_{s,k}^{\text{emb}} \|f\|_{H^s},$$

where

$$C_{s,k}^{\text{emb}} := (2\pi)^{-3/2} \max_{0 \leq j \leq k} \left(\int_{\mathbb{R}^3} |\xi|^{2j} \langle \xi \rangle^{-2s} d\xi \right)^{1/2} < \infty.$$

In particular,

$$H^\infty(\mathbb{R}^3) \hookrightarrow C_b^\infty(\mathbb{R}^3).$$

Proof. The first assertion follows from $\langle \xi \rangle^{2s_1} \leq \langle \xi \rangle^{2s_2}$. For $|\alpha| \leq k$, Fourier inversion and Cauchy–Schwarz give

$$|\partial^\alpha f(x)| \leq (2\pi)^{-3/2} \left(\int_{\mathbb{R}^3} |\xi|^{2|\alpha|} \langle \xi \rangle^{-2s} d\xi \right)^{1/2} \|f\|_{H^s}.$$

The integral is finite exactly when $s > |\alpha| + 3/2$. Approximation by Schwartz functions and dominated convergence give continuous representatives. For each k , choose $s = k + 2$. $\square$

Lemma 1.3 (Sobolev multiplication). *For every $m \in \mathbb{N}_0$ there is $C_m^{\text{alg}} < \infty$ such that*

$$\|fg\|_{H^m} \leq C_m^{\text{alg}} \|f\|_{H^{m+1}} \|g\|_{H^{m+1}}.$$

For $m \geq 2$,

$$\|fg\|_{H^m} \leq C_m^{\text{alg}} \|f\|_{H^m} \|g\|_{H^m}.$$

Proof. For $m = 0$, interpolation between L^2 and L^6 yields

$$\|fg\|_2 \leq \|f\|_4 \|g\|_4 \leq C \|f\|_{H^1} \|g\|_{H^1}.$$

For $m \geq 1$, apply the Leibniz identity to every $\partial^\alpha(fg)$, $|\alpha| \leq m$. In each term place the factor with at most $\lfloor m/2 \rfloor$ derivatives in L^∞ , using Lemma 1.2, and the other factor in L^2 . This proves the first estimate. For $m = 2$, the terms in which one derivative falls on each factor satisfy

$$\|\nabla f \nabla g\|_{L^2} \leq \|\nabla f\|_{L^6} \|\nabla g\|_{L^3} \leq C \|f\|_{H^2} \|g\|_{H^2},$$

where $\|\nabla g\|_{L^3} \leq \|\nabla g\|_{L^2}^{1/2} \|\nabla g\|_{L^6}^{1/2}$. The remaining derivatives of fg are bounded by placing the undifferentiated factor in L^∞ . This proves the second estimate for $m = 2$. For $m \geq 3$, the preceding Leibniz argument uses $\lfloor m/2 \rfloor + 2 \leq m$ and proves the second estimate. $\square$

Lemma 1.4 (Riesz and Leray multipliers). *Let*

$$r_j(\xi) := \begin{cases} i\xi_j/|\xi|, & \xi \neq 0, \\ 0, & \xi = 0, \end{cases} \quad P(\xi) := \begin{cases} I - \frac{\xi \otimes \xi}{|\xi|^2}, & \xi \neq 0, \\ I, & \xi = 0, \end{cases}$$

and

$$\widehat{R_j f}(\xi) = r_j(\xi) \widehat{f}(\xi), \quad \widehat{\mathbb{P}f}(\xi) = P(\xi) \widehat{f}(\xi).$$

For all $s \in \mathbb{R}$,

$$\|R_j f\|_{H^s} \leq \|f\|_{H^s}, \quad \|\mathbb{P}f\|_{H^s} \leq \|f\|_{H^s}.$$

If $m \in \mathbb{N}_0$ and $v \in H_\sigma^{m+1}$, then

$$q := \sum_{i,j=1}^3 R_i R_j (v_i v_j) \in H^m, \quad -\Delta q = \sum_{i,j=1}^3 \partial_i \partial_j (v_i v_j),$$

and

$$\mathbb{P}\nabla \cdot (v \otimes v) = (v \cdot \nabla)v + \nabla q.$$

Here

$$(v \otimes v)_{ij} := v_i v_j, \quad (\nabla \cdot (v \otimes v))_i := \sum_{j=1}^3 \partial_j (v_i v_j), \quad ((v \cdot \nabla)v)_i := \sum_{j=1}^3 v_j \partial_j v_i.$$

Proof. The multiplier symbols have operator norm at most one. Plancherel proves the first two estimates. Lemma 1.3 gives $v_i v_j \in H^m$. The remaining identities follow after multiplication by $|\xi|^2$ and by the symbol of $\mathbb{P}$. $\square$

Lemma 1.5 (Mixed Leibniz identities). *For $N \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$\partial_t^N \partial_x^\alpha (fg) = \sum_{a=0}^N \sum_{\beta \leq \alpha} \binom{N}{a} \binom{\alpha}{\beta} (\partial_t^a \partial_x^\beta f) (\partial_t^{N-a} \partial_x^{\alpha-\beta} g).$$

Proof. Apply the one-variable Leibniz identity successively in t, x_1, x_2, x_3 . $\square$

Lemma 1.6 (Adjacent Hilbert-scale trace). *Let $0 < T < \infty$, $m \in \mathbb{R}$, and*

$$v \in L^2(0, T; H^{m+1}), \quad \partial_t v \in L^2(0, T; H^{m-1}).$$

Then v has a unique representative in $C([0, T]; H^m)$, and

$$\|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 = 2 \operatorname{Re} \int_r^t \langle A^{m-1} \partial_t v(s), A^{m+1} v(s) \rangle_{L^2} ds, \quad (1)$$

$$\left| \|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 \right| \leq 2 \|\partial_t v\|_{L^2(r, t; H^{m-1})} \|v\|_{L^2(r, t; H^{m+1})}, \quad (2)$$

$$\sup_{0 \leq t \leq T} \|v(t)\|_{H^m}^2 \leq T^{-1} \|v\|_{L^2(0, T; H^m)}^2 + 2 \|\partial_t v\|_{L^2(0, T; H^{m-1})} \|v\|_{L^2(0, T; H^{m+1})}. \quad (3)$$

Proof. Let S_R denote Fourier restriction to $\{|\xi| \leq R\}$. For $v_R = S_R v$,

$$\frac{d}{dt} \|v_R(t)\|_{H^m}^2 = 2 \operatorname{Re} \langle A^{m-1} \partial_t v_R(t), A^{m+1} v_R(t) \rangle_{L^2}.$$

Integration and Cauchy–Schwarz prove (1)–(2) for v_R . Applied to $S_R v - S_S v$, the same estimate shows that $(v_R)_R$ is Cauchy in $C([0, T]; H^m)$; choose one Lebesgue time for the initial Cauchy term. Its limit equals v in $L^2(0, T; H^m)$. Passage to the limit proves the first two formulas. Choose $r \in (0, T)$ with $\|v(r)\|_{H^m}^2 \leq T^{-1} \|v\|_{L^2 H^m}^2$ and apply (2). $\square$

Lemma 1.7 (Frequency tail). *If $s > \sigma$, $K \geq 1$, and $f \in \dot{H}^s(\mathbb{R}^3)$, then*

$$\int_{|\xi| > K} |\xi|^{2\sigma} \left| \widehat{f}(\xi) \right|^2 d\xi \leq K^{-2(s-\sigma)} \|f\|_{\dot{H}^s}^2.$$

Proof. On $\{|\xi| > K\}$, $|\xi|^{2\sigma} \leq K^{-2(s-\sigma)} |\xi|^{2s}$. □

Lemma 1.8 (Integer transport commutator). *For every integer $m \geq 3$, there is $C_m^{\text{com}} < \infty$ such that every $v \in H_\sigma^{m+1}(\mathbb{R}^3)$ satisfies*

$$|\operatorname{Re} \langle A^m \mathbb{P} \nabla \cdot (v \otimes v), A^m v \rangle_{L^2}| \leq C_m^{\text{com}} \|\nabla v\|_{L^\infty} \|v\|_{H^m}^2. \quad (4)$$

Proof. For $|\alpha| \leq m$, set

$$\alpha! := \alpha_1! \alpha_2! \alpha_3!, \quad c_{m,\alpha} := \frac{m!}{(m - |\alpha|)! \alpha!}.$$

The multiplier $A^{2m} = (1 - \Delta)^m$ and the multinomial identity give the exact pairing formula

$$\langle A^m f, A^m g \rangle_{L^2} = \sum_{|\alpha| \leq m} c_{m,\alpha} \langle \partial^\alpha f, \partial^\alpha g \rangle_{L^2}. \quad (5)$$

The Leray projection is self-adjoint, commutes with A^m , and fixes $A^m v$. Since $\nabla \cdot (v \otimes v) = (v \cdot \nabla) v$, it follows that

$$\langle A^m \mathbb{P} \nabla \cdot (v \otimes v), A^m v \rangle_{L^2} = \sum_{|\alpha| \leq m} c_{m,\alpha} \langle \partial^\alpha ((v \cdot \nabla) v), \partial^\alpha v \rangle_{L^2}. \quad (6)$$

For each α ,

$$\partial^\alpha ((v \cdot \nabla) v) = (v \cdot \nabla) \partial^\alpha v + \sum_{0 < \beta \leq \alpha} \binom{\alpha}{\beta} (\partial^\beta v \cdot \nabla) \partial^{\alpha-\beta} v.$$

The first term has zero real pairing with $\partial^\alpha v$, because

$$\operatorname{Re} \langle (v \cdot \nabla) \partial^\alpha v, \partial^\alpha v \rangle_{L^2} = \frac{1}{2} \int_{\mathbb{R}^3} v \cdot \nabla |\partial^\alpha v|^2 dx = 0.$$

For a remaining term, put $\ell = |\alpha|$ and $k = |\beta|$. When $\ell \geq 2$, apply the Gagliardo–Nirenberg inequalities to $f = \nabla v$, with $M = \ell - 1$, $j = k - 1$, and the endpoint conventions at $j = 0, M$:

$$\|D^j f\|_{L^{2M/j}} \leq C_M \|f\|_{L^\infty}^{1-j/M} \|D^M f\|_{L^2}^{j/M}.$$

Here $L^{2M/0}$ means L^∞ . Hölder's inequality with the exponents belonging to j and $M - j$, whose reciprocal sum is $1/2$, yields

$$\left\| (\partial^\beta v \cdot \nabla) \partial^{\alpha-\beta} v \right\|_{L^2} \leq C_m \|\nabla v\|_{L^\infty} \|v\|_{H^\ell}.$$

The same estimate for $\ell = 1$ is the direct $L^\infty \cdot L^2$ bound. Multiplication by $\|\partial^\alpha v\|_2$, summation over β, α , and (6) prove (4). Approximation by divergence-free Schwartz fields justifies the integrations by parts for $v \in H_\sigma^{m+1}$. □

Theorem 1.9 (Quantitative local restart). *Set $C_q := \sqrt{3} C_3^{\text{alg}}$. For $\nu > 0$, set*

$$L_\nu := 4(1 + \nu + \nu^{-1}) \exp\left(\frac{\nu + \nu^{-1}}{2}\right), \quad c_\nu := (8C_q L_\nu^2)^{-2},$$

Then every $a \in H_\sigma^3(\mathbb{R}^3)$ generates a unique solution

$$w \in C([0, \delta]; H_\sigma^3) \cap L^2(0, \delta; H^4), \quad w_t \in L^2(0, \delta; H^2),$$

for every

$$0 < \delta \leq \delta_\nu(a) := \min\{1, c_\nu(1 + \|a\|_{H^3})^{-2}\}.$$

Its normalized pressure is $q = R_i R_j(w_i w_j)$. If $a \in H_\sigma^\infty$, then

$$\partial_t^n w, \partial_t^n q \in C([0, \delta]; H^m) \quad (n, m \in \mathbb{N}_0).$$

Proof. Let $0 < \delta \leq 1$, let

$$z_t - \nu \Delta z = F, \quad z(0) = a, \quad a \in H_\sigma^3, \quad F \in L^2(0, \delta; H_\sigma^2),$$

and set

$$A_0 := \|a\|_{H^3}, \quad G := \|F\|_{L^2(0, \delta; H^2)}.$$

The H^3 energy identity is

$$\frac{1}{2} \frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 = \text{Re} \langle A^2 F, A^4 z \rangle_{L^2}.$$

Since $\|z\|_{H^4}^2 = \|z\|_{H^3}^2 + \|\nabla z\|_{H^3}^2$, Young's inequality gives

$$\frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 \leq \nu \|z\|_{H^3}^2 + \nu^{-1} \|F\|_{H^2}^2.$$

Consequently,

$$\|z\|_{C([0, \delta]; H^3)} \leq e^{\nu/2} (A_0 + \nu^{-1/2} G). \quad (7)$$

Pairing $A^2 z_t - \nu \Delta A^2 z = A^2 F$ with $-\Delta A^2 z$ gives

$$\frac{1}{2} \frac{d}{dt} \|\nabla A^2 z\|_2^2 + \nu \|\Delta A^2 z\|_2^2 = \text{Re} \langle A^2 F, -\Delta A^2 z \rangle_{L^2}.$$

Integration and Young's inequality yield

$$\|\Delta z\|_{L^2(0, \delta; H^2)} \leq \nu^{-1/2} A_0 + \nu^{-1} G. \quad (8)$$

The pointwise multiplier inequality

$$\langle \xi \rangle^8 \leq 2(\langle \xi \rangle^6 + |\xi|^4 \langle \xi \rangle^4)$$

and $\delta \leq 1$ imply

$$\|z\|_{L^2(0, \delta; H^4)} \leq \sqrt{2} \left(\|z\|_{C([0, \delta]; H^3)} + \|\Delta z\|_{L^2(0, \delta; H^2)} \right). \quad (9)$$

The equation and (8) give

$$\|z_t\|_{L^2(0, \delta; H^2)} \leq \nu^{1/2} A_0 + 2G. \quad (10)$$

Equations (7)–(10) give

$$\begin{aligned} & \|z\|_{CH^3} + \|z\|_{L^2H^4} + \|z_t\|_{L^2H^2} \\ & \leq e^{\nu/2}(A_0 + \nu^{-1/2}G) + \sqrt{2} \left(e^{\nu/2}(A_0 + \nu^{-1/2}G) + \nu^{-1/2}A_0 + \nu^{-1}G \right) + \nu^{1/2}A_0 + 2G \\ & \leq L_\nu(A_0 + G). \end{aligned}$$

The last inequality follows from

$$\nu^{1/2} \leq \frac{1+\nu}{2}, \quad \nu^{-1/2} \leq \frac{1+\nu^{-1}}{2}, \quad e^{\nu/2} \leq e^{(\nu+\nu^{-1})/2}.$$

Fourier truncation and passage to the limit justify all four linear identities for the stated data class.

For $v, z \in H_\sigma^3$, Lemmas 1.3 and 1.4 give

$$\|\mathbb{P}\nabla \cdot (v \otimes z)\|_{H^2} \leq C_q \|v\|_{H^3} \|z\|_{H^3}, \quad (11)$$

$$\|\mathbb{P}\nabla \cdot (v \otimes v - z \otimes z)\|_{H^2} \leq C_q (\|v\|_{H^3} + \|z\|_{H^3}) \|v - z\|_{H^3}. \quad (12)$$

Indeed,

$$\|\nabla \cdot (v \otimes z)\|_{H^2}^2 \leq 3 \sum_{i,j=1}^3 \|v_i z_j\|_{H^3}^2 \leq 3(C_3^{\text{alg}})^2 \|v\|_{H^3}^2 \|z\|_{H^3}^2.$$

Define

$$X_\delta := \{v \in C([0, \delta]; H_\sigma^3) \cap L^2(0, \delta; H_\sigma^4) : v_t \in L^2(0, \delta; H_\sigma^2)\},$$

$$\|v\|_{X_\delta} := \|v\|_{CH^3} + \|v\|_{L^2H^4} + \|v_t\|_{L^2H^2}, \quad X_\delta(a) := \{v \in X_\delta : v(0) = a\}.$$

X_δ is a Banach space with this norm, and $X_\delta(a)$ is closed. For $A_0 > 0$, set $R := 2L_\nu A_0$ and

$$\mathbb{B}_R := \{v \in X_\delta(a) : \|v\|_{X_\delta} \leq R\}.$$

For $v \in \mathbb{B}_R$, define

$$\Phi(v)(t) := e^{\nu t \Delta} a - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P}\nabla \cdot (v \otimes v)(s) \, ds.$$

If $\delta \leq \delta_\nu(a)$, then

$$\delta^{1/2} \leq \frac{1}{8C_q L_\nu^2 (1 + A_0)}.$$

The linear estimate and (11) imply

$$\begin{aligned} \|\Phi(v)\|_{X_\delta} & \leq L_\nu (A_0 + C_q \delta^{1/2} R^2) \\ & \leq L_\nu A_0 + \frac{L_\nu A_0^2}{2(1 + A_0)} < R. \end{aligned}$$

For $v, z \in \mathbb{B}_R$,

$$\begin{aligned} \|\Phi(v) - \Phi(z)\|_{X_\delta} & \leq L_\nu C_q \delta^{1/2} (\|v\|_{CH^3} + \|z\|_{CH^3}) \|v - z\|_{CH^3} \\ & \leq \frac{A_0}{2(1 + A_0)} \|v - z\|_{X_\delta} \leq \frac{1}{2} \|v - z\|_{X_\delta}. \end{aligned}$$

Thus Φ has a unique fixed point in $\mathbb{B}_R$. If $A_0 = 0$, the zero function is the fixed point. For any two mild solutions $v, z \in C([0, \delta]; H_\sigma^3)$, set

$$M := \|v\|_{CH^3} + \|z\|_{CH^3}.$$

On every subinterval of length $h \leq 1$ on which the two solutions have the same initial value, the zero-data linear estimate gives

$$\|v - z\|_{CH^3} \leq L_\nu C_q h^{1/2} M \|v - z\|_{CH^3}.$$

A finite partition with $L_\nu C_q h^{1/2} M < 1$ proves uniqueness in the full class $C([0, \delta]; H_\sigma^3)$. For the fixed point, (11) places the forcing in $L^2(0, \delta; H^2)$; the linear estimates give

$$w \in L^2(0, \delta; H^4), \quad w_t \in L^2(0, \delta; H^2).$$

Lemma 1.4 gives the normalized pressure $q = R_i R_j (w_i w_j)$.

For $m \geq 3$, Lemma 1.8 gives

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 \leq C_m^{\text{com}} \|\nabla w\|_{L^\infty} \|w\|_{H^m}^2.$$

Since $w \in CH^3 \hookrightarrow CW^{1,\infty}$, Gronwall's inequality and Fourier-cutoff approximation give, on the same interval,

$$a \in H_\sigma^m \implies w \in CH_\sigma^m \cap L^2 H_\sigma^{m+1}, \quad w_t \in L^2 H_\sigma^{m-1}.$$

Assume now that $a \in H_\sigma^\infty$. Set $W_0 = w$ and define recursively

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} W_{a,i} W_{n-a,j},$$

$$W_{n+1} = \nu \Delta W_n - \sum_{a=0}^n \binom{n}{a} (W_a \cdot \nabla) W_{n-a} - \nabla Q_n.$$

If $W_0, \dots, W_n \in CH^{m+2}$, then Lemmas 1.3 and 1.4 give

$$Q_n \in CH^{m+1}, \quad W_{n+1} \in CH^m.$$

Since m is arbitrary, induction gives $W_n, Q_n \in CH^m$ for every $n, m \in \mathbb{N}_0$. The equation first gives $W_0 \in C^1 H^m$ and $\partial_t W_0 = W_1$. If $W_a = \partial_t^a w$ for $0 \leq a \leq n$, distributional differentiation of $q = R_i R_j (w_i w_j)$ and Lemma 1.5 give $\partial_t^n q = Q_n$. Differentiation of the momentum equation then gives $\partial_t W_n = W_{n+1}$. Induction gives

$$W_n \in C^1 H^m, \quad \partial_t W_n = W_{n+1}, \quad W_n = \partial_t^n w, \quad Q_n = \partial_t^n q.$$

□

2 The pressure-complete setting and mixed jet

Proposition 2.1 (Pressure normalization). *Let u, p be smooth on $I \times \mathbb{R}^3$, let $\nabla \cdot u = 0$, and suppose*

$$u(t) \in H_\sigma^1(\mathbb{R}^3), \quad p(t) \in H^0(\mathbb{R}^3) \quad (t \in I),$$

and

$$u_t + (u \cdot \nabla)u + \nabla p = \nu \Delta u. \tag{13}$$

Then

$$-\Delta p = \partial_i \partial_j (u_i u_j), \tag{14}$$

$$p = R_i R_j (u_i u_j), \tag{15}$$

$$(u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u). \tag{16}$$

The condition $p(t) \in H^0(\mathbb{R}^3)$ fixes the pressure uniquely.

Proof. Incompressibility gives

$$(u \cdot \nabla)u_i = \partial_j(u_j u_i).$$

Taking the divergence of (13) cancels $\nabla \cdot u_t$ and $\nabla \cdot \Delta u$, which yields (14). Lemma 1.4, applied with $m = 0$, gives $q := R_i R_j(u_i u_j) \in H^0$, the same Poisson identity, and

$$(u \cdot \nabla)u + \nabla q = \mathbb{P} \nabla \cdot (u \otimes u).$$

If $p_1, p_2 \in H^0$ satisfy (14), then $h = p_1 - p_2$ satisfies $|\xi|^2 \widehat{h} = 0$. Since $\widehat{h} \in L^2$ and is supported at $\{0\}$, it vanishes. Hence $p = q$, which proves (15) and (16). $\square$

Definition 2.2 (Maximal pressure-complete classical history). Fix $\nu > 0$ and $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$. A pressure-complete classical history generated by (u_0, ν) is a triple

$$(T_*, u, p), \quad 0 < T_* \leq \infty,$$

such that, for every $0 < T < T_*$ and $n, m \in \mathbb{N}_0$,

$$\partial_t^n u \in C([0, T]; H_\sigma^m(\mathbb{R}^3)),$$

$$\partial_t^n p \in C([0, T]; H^m(\mathbb{R}^3)),$$

and

$$u_t + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \tag{17}$$

$$\nabla \cdot u = 0, \tag{18}$$

$$u(0) = u_0, \tag{19}$$

$$p(t) = R_i R_j(u_i(t) u_j(t)). \tag{20}$$

The history is maximal if no pressure-complete classical history $(\widetilde{T}, \widetilde{u}, \widetilde{p})$ generated by (u_0, ν) satisfies

$$\widetilde{T} > T_*, \quad (\widetilde{u}, \widetilde{p})|_{[0, T_*]} = (u, p).$$

Definition 2.3 (Temporal and mixed velocity–pressure jets). Let (T_*, u, p) be a maximal pressure-complete classical history. For $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$, define

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad J_{n, \alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n, \alpha} := \partial_t^n \partial_x^\alpha p.$$

For $\beta \leq \alpha$, define

$$\binom{\alpha}{\beta} := \prod_{\ell=1}^3 \binom{\alpha_\ell}{\beta_\ell}, \quad |\alpha| := \alpha_1 + \alpha_2 + \alpha_3.$$

Proposition 2.4 (Pressure-complete binomial recurrences). For every $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a, i} V_{n-a, j}, \tag{21}$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n, \tag{22}$$

$$\Pi_{n, \alpha} = R_i R_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} J_{a, \beta, i} J_{n-a, \alpha-\beta, j}, \tag{23}$$

$$J_{n+1, \alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a, \beta} \cdot \nabla) J_{n-a, \alpha-\beta} + \nabla \Pi_{n, \alpha} = \nu \Delta J_{n, \alpha}, \tag{24}$$

$$\nabla \cdot J_{n, \alpha} = 0. \tag{25}$$

Proof. Apply ∂_t^n to (20) and use Lemma 1.5 with $\alpha = 0$ to obtain (21). Apply ∂_t^n to (17) to obtain (22). Apply ∂_x^α to both identities, use Lemma 1.5, and commute the derivatives with $R_i R_j$ to obtain (23) and (24). Differentiation of (18) gives (25). $\square$

Proposition 2.5 (Mixed-index transport matrix and pressure completion). *Let*

$$\Gamma := \mathbb{N}_0 \times \mathbb{N}_0^3, \quad \kappa = (n, \alpha), \quad \lambda = (a, \beta),$$

and write

$$\lambda \preceq \kappa \iff a \leq n \text{ and } \beta \leq \alpha, \quad \kappa - \lambda := (n - a, \alpha - \beta), \quad \binom{\kappa}{\lambda} := \binom{n}{a} \binom{\alpha}{\beta}.$$

Set

$$J_\kappa := J_{n,\alpha}, \quad \Pi_\kappa := \Pi_{n,\alpha}, \quad \mathbf{J} := (J_\lambda)_{\lambda \in \Gamma}.$$

The block operator

$$\mathbb{L}(J)_{\kappa,\lambda} := \begin{cases} \binom{\kappa}{\lambda} \mathbb{P}[(J_{\kappa-\lambda} \cdot \nabla) \cdot], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa, \end{cases} \quad (26)$$

has the finite block-row action

$$[\mathbb{L}(J)\mathbf{J}]_\kappa = \sum_{\lambda \preceq \kappa} \binom{\kappa}{\lambda} \mathbb{P}[(J_\lambda \cdot \nabla) J_{\kappa-\lambda}]. \quad (27)$$

Consequently the complete mixed row is the pair

$$J_{n+1,\alpha} + [\mathbb{L}(J)\mathbf{J}]_\kappa = \nu \Delta J_\kappa, \quad (28)$$

$$\Pi_\kappa = R_i R_j \sum_{\lambda \preceq \kappa} \binom{\kappa}{\lambda} J_{\lambda,i} J_{\kappa-\lambda,j}, \quad (29)$$

while the unprojected velocity identity retains $\nabla \Pi_\kappa$ exactly as in (24).

Order Γ by increasing $n + |\alpha|$, with any fixed ordering among indices of equal depth. Then $\mathbb{L}(J)$ is block lower triangular. For

$$\kappa! := n! \alpha!, \quad \alpha! := \alpha_1! \alpha_2! \alpha_3!, \quad \hat{J}_\kappa := \frac{J_\kappa}{\kappa!}, \quad \hat{\Pi}_\kappa := \frac{\Pi_\kappa}{\kappa!},$$

let D be the diagonal matrix with $D_{\kappa,\kappa} = \kappa!$. On every finite downward-closed truncation of Γ ,

$$D^{-1} \mathbb{L}(D\hat{J})D = \hat{\mathbb{L}}(\hat{J}), \quad \hat{\mathbb{L}}(\hat{J})_{\kappa,\lambda} := \begin{cases} \mathbb{P}[(\hat{J}_{\kappa-\lambda} \cdot \nabla) \cdot], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa. \end{cases} \quad (30)$$

Thus the factorial-normalized pressure-complete recurrence is

$$(n+1)\hat{J}_{n+1,\alpha} + \sum_{\lambda \preceq \kappa} (\hat{J}_\lambda \cdot \nabla) \hat{J}_{\kappa-\lambda} + \nabla \hat{\Pi}_\kappa = \nu \Delta \hat{J}_\kappa, \quad (31)$$

$$\hat{\Pi}_\kappa = R_i R_j \sum_{\lambda \preceq \kappa} \hat{J}_{\lambda,i} \hat{J}_{\kappa-\lambda,j}. \quad (32)$$

The normalized transport blocks depend on the difference $\kappa - \lambda$; hence (30) is the block lower-triangular convolution form of the mixed transport array.

Proof. For $\lambda \preceq \kappa$, change the summation index from λ to $\kappa - \lambda$ and use $\binom{\kappa}{\kappa-\lambda} = \binom{\kappa}{\lambda}$. This gives (27). Apply $\mathbb{P}$ to (24). The identities

$$\mathbb{P}J_{n+1,\alpha} = J_{n+1,\alpha}, \quad \mathbb{P}\Delta J_\kappa = \Delta J_\kappa, \quad \mathbb{P}\nabla\Pi_\kappa = 0$$

follow from (25) and the Fourier symbols. They prove (28); equation (29) is (23) in mixed-index notation. The pressure formula also shows that projection of the velocity row loses no pressure coordinate.

If $\lambda \preceq \kappa$, then $a + |\beta| \leq n + |\alpha|$, which proves lower triangularity in the stated ordering. Finally,

$$\binom{\kappa}{\lambda} \lambda! (\kappa - \lambda)! = \kappa!.$$

Dividing the mixed velocity and pressure rows by $\kappa!$ proves (31) and (32). The same factorial cancellation in each operator block proves (30). Every row has $(n+1) \prod_{j=1}^3 (\alpha_j + 1)$ supported blocks, so all displayed row sums are finite. $\square$

Definition 2.6 (Canonical transport and pressure constants). For vector fields f, g , set

$$\mathcal{B}(f, g) := (f \cdot \nabla)g, \quad \mathcal{Q}(f, g) := \sum_{i,j=1}^3 R_i R_j (f_i g_j).$$

For $m \in \mathbb{N}_0$, define

$$C_m^{\text{tr}} := \sup_{\substack{f \in H^{m+1}(\mathbb{R}^3; \mathbb{R}^3) \setminus \{0\} \\ g \in H^{m+2}(\mathbb{R}^3; \mathbb{R}^3) \setminus \{0\}}} \frac{\|\mathcal{B}(f, g)\|_{H^m}}{\|f\|_{H^{m+1}} \|g\|_{H^{m+2}}},$$

$$C_m^{\text{p}} := \sup_{f, g \in H^{m+1}(\mathbb{R}^3; \mathbb{R}^3) \setminus \{0\}} \frac{\|\mathcal{Q}(f, g)\|_{H^m} + \|\nabla \mathcal{Q}(f, g)\|_{H^{m-1}}}{\|f\|_{H^{m+1}} \|g\|_{H^{m+1}}}.$$

Lemma 2.7. For every $m \in \mathbb{N}_0$,

$$C_m^{\text{tr}} < \infty, \quad C_m^{\text{p}} < \infty.$$

Proof. Lemma 1.3 and componentwise Cauchy–Schwarz give

$$\|\mathcal{B}(f, g)\|_{H^m} \leq C_m^{\text{alg}} \|f\|_{H^{m+1}} \|g\|_{H^{m+2}}.$$

The same lemma, the multiplier bounds in Lemma 1.4, and $\|\nabla h\|_{H^{m-1}} \leq \|h\|_{H^m}$ give

$$\|\mathcal{Q}(f, g)\|_{H^m} + \|\nabla \mathcal{Q}(f, g)\|_{H^{m-1}} \leq 6C_m^{\text{alg}} \|f\|_{H^{m+1}} \|g\|_{H^{m+1}}.$$

$\square$

Lemma 2.8 (Initial pressure-complete jet). Set

$$V_n^0 := V_n(0), \quad Q_n^0 := Q_n(0), \quad J_{n,\alpha}^0 := J_{n,\alpha}(0), \quad \Pi_{n,\alpha}^0 := \Pi_{n,\alpha}(0).$$

Then

$$V_0^0 = u_0, \tag{33}$$

$$Q_n^0 = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^0 V_{n-a,j}^0, \tag{34}$$

$$V_{n+1}^0 = \nu \Delta V_n^0 - \sum_{a=0}^n \binom{n}{a} (V_a^0 \cdot \nabla) V_{n-a}^0 - \nabla Q_n^0. \quad (35)$$

For every $n, m \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,

$$V_n^0, Q_n^0, J_{n,\alpha}^0, \Pi_{n,\alpha}^0 \in H^m(\mathbb{R}^3).$$

Moreover,

$$\|Q_n^0\|_{H^m} \leq C_m^p \sum_{a=0}^n \binom{n}{a} \|V_a^0\|_{H^{m+1}} \|V_{n-a}^0\|_{H^{m+1}}, \quad (36)$$

$$\|V_{n+1}^0\|_{H^m} \leq \nu \|V_n^0\|_{H^{m+2}} + C_m^{\text{tr}} \sum_{a=0}^n \binom{n}{a} \|V_a^0\|_{H^{m+1}} \|V_{n-a}^0\|_{H^{m+2}} + \|Q_n^0\|_{H^{m+1}}. \quad (37)$$

For $m \in \mathbb{N}_0$, set

$$\mathcal{U}_m(u_0)^2 := \sum_{\substack{\gamma \in \mathbb{N}_0^3 \\ |\gamma| \leq m}} \frac{m!}{(m-|\gamma|)! \gamma!} \|\partial^\gamma u_0\|_{L^2}^2, \quad \gamma! := \gamma_1! \gamma_2! \gamma_3!,$$

and define $D_{n,m} = D_{n,m}(u_0, \nu)$ and $E_{n,m} = E_{n,m}(u_0, \nu)$ recursively by

$$D_{0,m} := \mathcal{U}_m(u_0), \quad (38)$$

$$E_{n,m} := C_m^p \sum_{a=0}^n \binom{n}{a} D_{a,m+1} D_{n-a,m+1}, \quad (39)$$

$$D_{n+1,m} := \nu D_{n,m+2} + C_m^{\text{tr}} \sum_{a=0}^n \binom{n}{a} D_{a,m+1} D_{n-a,m+2} + E_{n,m+1}. \quad (40)$$

Then

$$\|V_n^0\|_{H^m} \leq D_{n,m}, \quad \|Q_n^0\|_{H^m} \leq E_{n,m}, \quad (41)$$

$$\|J_{n,\alpha}^0\|_{H^m} \leq D_{n,m+|\alpha|}, \quad \|\Pi_{n,\alpha}^0\|_{H^m} \leq E_{n,m+|\alpha|}. \quad (42)$$

For fixed n, m , $D_{n,m}$ depends only on

$$\nu, \quad \{\|\partial^\gamma u_0\|_{L^2} : |\gamma| \leq m + 2n\},$$

and $E_{n,m}$ depends only on

$$\nu, \quad \{\|\partial^\gamma u_0\|_{L^2} : |\gamma| \leq m + 2n + 1\}.$$

Proof. Equations (33)–(35) are (21) and (22) at $t = 0$. Assume $V_0^0, \dots, V_n^0 \in \bigcap_{m \geq 0} H^m$. Lemmas 1.3 and 1.4 give

$$Q_n^0 \in \bigcap_{m \geq 0} H^m, \quad V_{n+1}^0 \in \bigcap_{m \geq 0} H^m.$$

Induction starts from $u_0 \in \mathcal{S}_\sigma$. Spatial differentiation and (23) give the claims for $J_{n,\alpha}^0$ and $\Pi_{n,\alpha}^0$.

The definition of C_m^p applied to (34) proves (36). The definition of C_m^{tr} , the inequality $\|\nabla h\|_{H^m} \leq \|h\|_{H^{m+1}}$, and (35) prove (37).

The multinomial identity

$$(1 + |\xi|^2)^m = \sum_{\substack{\gamma \in \mathbb{N}_0^3 \\ |\gamma| \leq m}} \frac{m!}{(m - |\gamma|)! \gamma!} \xi^{2\gamma}$$

and Plancherel give

$$\mathcal{U}_m(u_0) = \|u_0\|_{H^m}. \quad (43)$$

Equations (36) and (37), followed by induction on n , give (41). For every $\alpha \in \mathbb{N}_0^3$,

$$\|\partial^\alpha f\|_{H^m} \leq \|f\|_{H^{m+|\alpha|}}.$$

Since $J_{n,\alpha}^0 = \partial^\alpha V_n^0$ and $\Pi_{n,\alpha}^0 = \partial^\alpha Q_n^0$, this proves (42). A second induction on n in (39)–(40) gives the stated maximal derivative orders of u_0 . $\square$

Theorem 2.9 (Finite-row pressure estimates). *Let $I = (0, T) \subset (0, T_*)$, let $n, M \in \mathbb{N}_0$, and suppose*

$$V_a \in L^2(I; H^{M+1}(\mathbb{R}^3)) \quad (0 \leq a \leq n).$$

Then

$$\begin{aligned} & \|Q_n\|_{L^1(I; H^M)} + \|\nabla Q_n\|_{L^1(I; H^{M-1})} \\ & \leq C_M^p \sum_{a=0}^n \binom{n}{a} \|V_a\|_{L^2(I; H^{M+1})} \|V_{n-a}\|_{L^2(I; H^{M+1})}. \end{aligned} \quad (44)$$

If

$$\mathcal{V}_{n,M}(I)^2 := \sum_{a=0}^n \|V_a\|_{L^2(I; H^{M+1})}^2,$$

then

$$\|Q_n\|_{L^1(I; H^M)} + \|\nabla Q_n\|_{L^1(I; H^{M-1})} \leq C_M^p 2^n \mathcal{V}_{n,M}(I)^2. \quad (45)$$

For $\alpha \in \mathbb{N}_0^3$, assume

$$J_{a,\beta} \in L^2(I; H^{M+1}(\mathbb{R}^3)) \quad (0 \leq a \leq n, \beta \leq \alpha).$$

Then

$$\begin{aligned} & \|\Pi_{n,\alpha}\|_{L^1(I; H^M)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I; H^{M-1})} \\ & \leq C_M^p \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} \|J_{a,\beta}\|_{L^2(I; H^{M+1})} \|J_{n-a,\alpha-\beta}\|_{L^2(I; H^{M+1})}. \end{aligned} \quad (46)$$

If

$$\mathcal{J}_{n,\alpha,M}(I)^2 := \sum_{a=0}^n \sum_{\beta \leq \alpha} \|J_{a,\beta}\|_{L^2(I; H^{M+1})}^2,$$

then

$$\|\Pi_{n,\alpha}\|_{L^1(I; H^M)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I; H^{M-1})} \leq C_M^p 2^{n+|\alpha|} \mathcal{J}_{n,\alpha,M}(I)^2. \quad (47)$$

Proof. Lemmas 1.3 and 1.4 give

$$\|R_i R_j(f_i g_j)\|_{H^M} + \|\nabla R_i R_j(f_i g_j)\|_{H^{M-1}} \leq C_M^p \|f\|_{H^{M+1}} \|g\|_{H^{M+1}}. \quad (48)$$

Equations (21) and (48) give, for almost every $t \in I$,

$$\|Q_n(t)\|_{H^M} + \|\nabla Q_n(t)\|_{H^{M-1}} \leq C_M^p \sum_{a=0}^n \binom{n}{a} \|V_a(t)\|_{H^{M+1}} \|V_{n-a}(t)\|_{H^{M+1}}.$$

Integration and Cauchy–Schwarz prove (44). For nonnegative x_a ,

$$\sum_{a=0}^n \binom{n}{a} x_a x_{n-a} \leq \sum_{a=0}^n \binom{n}{a} x_a^2 \leq 2^n \sum_{a=0}^n x_a^2.$$

This proves (45).

Apply (48) to each summand in (23). Integration and Cauchy–Schwarz give (46). The involution

$$(a, \beta) \mapsto (n - a, \alpha - \beta)$$

preserves the weights $\binom{n}{a} \binom{\alpha}{\beta}$, and

$$\sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} = 2^{n+|\alpha|}.$$

In particular, every individual weight is at most $2^{n+|\alpha|}$. The inequality $2xy \leq x^2 + y^2$ proves (47). $\square$

3 Datum-generated compatible rectangles

Proposition 3.1 (Kinetic-energy identity). *For every $0 \leq t < T_*$,*

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds = \frac{1}{2} \|u_0\|_2^2. \quad (49)$$

Consequently, for $0 < T < T_$,*

$$\|u\|_{L^\infty(0,T;L^2)} \leq \|u_0\|_2, \quad (50)$$

$$\|u\|_{L^2(0,T;H^1)}^2 \leq \left(T + \frac{1}{2\nu}\right) \|u_0\|_2^2. \quad (51)$$

Proof. Take the L^2 pairing of the momentum equation with u . Integration by parts and $\nabla \cdot u = 0$ give

$$\langle (u \cdot \nabla)u, u \rangle_{L^2} = 0, \quad \langle \nabla p, u \rangle_{L^2} = 0, \quad \langle \Delta u, u \rangle_{L^2} = -\|\nabla u\|_2^2.$$

Integration in time proves (49). The two estimates follow by discarding the nonnegative dissipation and adding $\int_0^T \|u(s)\|_2^2 \, ds$. $\square$

Definition 3.2 (Finite mixed rectangle). Let $0 < T < T_*$ and $N, M \in \mathbb{N}_0$. Define

$$\mathfrak{R}_{N,M}(u; T) := \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2, \quad (52)$$

$$\mathfrak{Q}_{N,M}(p; T) := \sum_{n=0}^N \left(\|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})} \right). \quad (53)$$

Set

$$\mathcal{R}_{N,M}[u_0; T] := ((V_n|_{(0,T)})_{n=0}^N, (V_{n+1}|_{(0,T)})_{n=0}^N; (Q_n|_{(0,T)})_{n=0}^N).$$

Sobolev rectangle matrix and adjacent stencil. For $m \in \mathbb{Z}$, set

$$G_{n,m}(T) := \|V_n\|_{L^2(0,T;H^m)},$$

and, for $M \in \mathbb{N}_0$, set

$$P_{n,M}(T) := \|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})}.$$

The scalar norm matrix of the pressure-complete rectangle is

$$\mathbf{C}_{N,M}(T) := \begin{pmatrix} G_{0,M+1}(T) & G_{1,M-1}(T) & P_{0,M}(T) \\ G_{1,M+1}(T) & G_{2,M-1}(T) & P_{1,M}(T) \\ \vdots & \vdots & \vdots \\ G_{N,M+1}(T) & G_{N+1,M-1}(T) & P_{N,M}(T) \end{pmatrix}. \quad (54)$$

If $\mathbf{G}_{N,M}(T)$ denotes its first two columns, then

$$\|\mathbf{G}_{N,M}(T)\|_{\mathbb{F}}^2 = \mathfrak{R}_{N,M}(u; T), \quad \sum_{n=0}^N P_{n,M}(T) = \mathfrak{Q}_{N,M}(p; T). \quad (55)$$

For each $0 \leq n \leq N$, the first two entries in row n form the adjacent Hilbert-scale stencil

$$\left. \begin{array}{l} (n, M+1) : \quad V_n \in L^2(0, T; H^{M+1}), \\ (n+1, M-1) : \quad V_{n+1} = \partial_t V_n \in L^2(0, T; H^{M-1}) \end{array} \right\} \longrightarrow (n, M) : \quad V_n \in C([0, T]; H^M). \quad (56)$$

For every $m \in \mathbb{N}_0$, this Sobolev matrix measures the mixed-index array of Proposition 2.5 through the exact identity

$$G_{n,m}(T)^2 = \sum_{\substack{\alpha \in \mathbb{N}_0^3 \\ |\alpha| \leq m}} \frac{m!}{(m - |\alpha|)! \alpha!} \|J_{n,\alpha}\|_{L^2((0,T) \times \mathbb{R}^3)}^2. \quad (57)$$

The pressure entry in row n is rooted at $Q_n = \Pi_{n,0}$; its spatial rows $\partial_x^\alpha Q_n = \Pi_{n,\alpha}$ are fixed by (29).

Proof. The two identities in (55) are the definitions (52) and (53), arranged by temporal row. Lemma 1.6 gives (56). Finally, Plancherel and

$$(1 + |\xi|^2)^m = \sum_{\substack{\alpha \in \mathbb{N}_0^3 \\ |\alpha| \leq m}} \frac{m!}{(m - |\alpha|)! \alpha!} \xi^{2\alpha}$$

give (57). The pressure claims are the definitions of $Q_n, \Pi_{n,\alpha}$ and (29). These identities organize the finite coordinates; their whole-terminal bounds are supplied by Theorem 3.3 and Proposition 3.4. $\square$

Theorem 3.3 (Datum-generated whole-terminal rectangles). *Let $\nu > 0$, $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let (T_*, u, p) be the maximal pressure-complete classical history generated by (u_0, ν) . If $T_* < \infty$, then, for every $N, M \in \mathbb{N}_0$,*

$$\mathfrak{R}_{N,M}^*[u_0, \nu, T_*] := \sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty. \quad (58)$$

Equivalently, for every $n, m \in \mathbb{N}_0$,

$$V_n \in L^2(0, T_*; H^{m+1}), \quad V_{n+1} \in L^2(0, T_*; H^{m-1}). \quad (59)$$

Every coordinate in (59) is the corresponding distributional derivative of the single maximal history.

Proposition 3.4 (Pressure completion). *Under the hypotheses of Theorem 3.3,*

$$\mathfrak{Q}_{N,M}^*[u_0, \nu, T_*] := \sup_{0 < T < T_*} \mathfrak{Q}_{N,M}(p; T) \leq C_M^p (2^{N+1} - 1) \mathfrak{R}_{N,M}^*[u_0, \nu, T_*]. \quad (60)$$

For every $0 \leq n \leq N$, the identities

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (61)$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n \quad (62)$$

hold respectively in $L^1(0, T_*; H^M)$ and $L^1(0, T_*; H^{M-1})$, including when $M = 0$.

Proof. For every $0 < T < T_*$, Theorem 2.9 gives

$$\|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})} \leq C_M^p 2^n \mathfrak{R}_{N,M}^*[u_0, \nu, T_*].$$

Sum over $0 \leq n \leq N$, then let $T \uparrow T_*$. Monotone convergence in the nonnegative norm integrals proves (60). Since $\nabla \cdot V_a = 0$,

$$(V_a \cdot \nabla) V_{n-a} = \nabla \cdot (V_a \otimes V_{n-a}).$$

Lemma 1.3 and Cauchy–Schwarz in time give

$$\|V_a \otimes V_{n-a}\|_{L^1(0,T_*;H^M)} \leq C_M^{\text{alg}} \|V_a\|_{L^2(0,T_*;H^{M+1})} \|V_{n-a}\|_{L^2(0,T_*;H^{M+1})}.$$

Thus every transport term belongs to $L^1(0, T_*; H^{M-1})$, also for $M = 0$. Moreover,

$$\|\Delta V_n\|_{L^1 H^{M-1}} \leq T_*^{1/2} \|V_n\|_{L^2 H^{M+1}}.$$

Theorem 3.3 places V_{n+1} in $L^2(0, T_*; H^{M-1})$, hence in $L^1(0, T_*; H^{M-1})$. The strict-interval identities are identities of distributions; their whole-terminal representatives belong to the stated spaces. This proves the two identities on $(0, T_*)$. $\square$

Definition 3.5 (Finite-window spaces). For $N, M \in \mathbb{N}_0$, set

$$\mathcal{X}_{N,M} := \prod_{n=0}^N L^2(0, T_*; H^{M+1}) \times \prod_{n=0}^N L^2(0, T_*; H^{M-1}) \times \prod_{n=0}^N L^1(0, T_*; H^M).$$

The whole-terminal rectangle is the element

$$\mathcal{R}_{N,M}[u_0; T_*] := ((V_n)_{n=0}^N, (V_{n+1})_{n=0}^N; (Q_n)_{n=0}^N) \in \mathcal{X}_{N,M}. \quad (63)$$

Define its datum fibre by

$$\mathcal{D}_{N,M}[u_0; T_*] := \{\mathcal{R}_{N,M}[u_0; T_*]\} \subset \mathcal{X}_{N,M}. \quad (64)$$

If $N' \leq N$ and $M' \leq M$, let

$$\rho_{N',M'}^{N,M} : \mathcal{X}_{N,M} \longrightarrow \mathcal{X}_{N',M'}$$

delete the rows with index larger than N' and apply

$$H^{M+1} \hookrightarrow H^{M'+1}, \quad H^{M-1} \hookrightarrow H^{M'-1}, \quad H^M \hookrightarrow H^{M'}$$

to the retained coordinates.

Proposition 3.6 (Restriction compatibility). *For $N' \leq N$ and $M' \leq M$,*

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0; T_*] = \mathcal{R}_{N',M'}[u_0; T_*]. \quad (65)$$

If $(N'', M'') \leq (N', M') \leq (N, M)$, then

$$\rho_{N'',M''}^{N',M'} \rho_{N',M'}^{N,M} = \rho_{N'',M''}^{N,M}.$$

Thus the restrictions of the maps $\rho_{N',M'}^{N,M}$ to the datum fibres form an inverse system.

Proof. All retained entries are the same distributions $\partial_t^n u$, $\partial_t^{n+1} u$, and $\partial_t^n p$. The Sobolev embeddings preserve those distributions. Deletion of rows and composition of canonical embeddings are associative. $\square$

Definition 3.7 (Compatible intersection).

$$\mathcal{I}[u_0] := \varprojlim_{N,M} \mathcal{D}_{N,M}[u_0; T_*] \quad (66)$$

is the subset of $\prod_{N,M \in \mathbb{N}_0} \mathcal{D}_{N,M}[u_0; T_*]$ given by

$$\mathcal{I}[u_0] = \left\{ (X_{N,M})_{N,M} : \rho_{N',M'}^{N,M} X_{N,M} = X_{N',M'} \text{ whenever } N' \leq N, M' \leq M \right\}. \quad (67)$$

Proposition 3.8 (Mixed Sobolev realization). *The velocity coordinate of $\mathcal{I}[u_0]$ satisfies*

$$u \in \bigcap_{r,m \in \mathbb{N}_0} H^r(0, T_*; H^m(\mathbb{R}^3)). \quad (68)$$

For every $r, m \in \mathbb{N}_0$,

$$\sum_{n=0}^r \|\partial_t^n u\|_{L^2(0, T_*; H^m)}^2 \leq \mathfrak{R}_{r,m}^*[u_0, \nu, T_*]. \quad (69)$$

Proof. Fix r, m . The rectangle $(N, M) = (r, m)$ contains

$$\partial_t^n u \in L^2(0, T_*; H^{m+1}) \hookrightarrow L^2(0, T_*; H^m) \quad (0 \leq n \leq r).$$

These entries are the distributional time derivatives of u , so $u \in H^r(0, T_*; H^m)$. The norm bound follows from (58) and Sobolev nesting. $\square$

Definition 3.9 (Critical-height rows). For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t),$$

$$\mathcal{P}_s(t) := -\operatorname{Re} \langle \Lambda^s u(t), \Lambda^s((u \cdot \nabla)u + \nabla p)(t) \rangle_{L^2}.$$

Proposition 3.10 (Completed height recurrence). *For every $0 < T < T_*$, $s \geq 0$, and $t \in (0, T)$,*

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s, \quad (70)$$

and, for every $n \in \mathbb{N}$ and $t \in (0, T)$,

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}. \quad (71)$$

Consequently, for every $n \in \mathbb{N}$ and $t \in (0, T)$,

$$\frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n} = E_{n+1/2}. \quad (72)$$

Proof. Pair the momentum equation with $\Lambda^{2s}u$. Self-adjointness of Λ and $\langle \Lambda^s u, \Lambda^s \Delta u \rangle_{L^2} = -\|\Lambda^{s+1}u\|_2^2$ give (70). We first prove, for every $n \in \mathbb{N}$,

$$\partial_t^n E_s = (-2\nu)^n E_{s+n} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{s+j}. \quad (73)$$

For $n = 1$, this is (70). Suppose it holds at order n . Differentiation gives

$$\partial_t^{n+1} E_s = (-2\nu)^n \partial_t E_{s+n} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-j} \mathcal{P}_{s+j}.$$

Substitution of $\partial_t E_{s+n} = -2\nu E_{s+n+1} + \mathcal{P}_{s+n}$ yields

$$\partial_t^{n+1} E_s = (-2\nu)^{n+1} E_{s+n+1} + \sum_{j=0}^n (-2\nu)^j \partial_t^{n-j} \mathcal{P}_{s+j},$$

which is (73) at order $n + 1$. Taking $s = 1/2$ proves (71), and division by $(-2\nu)^n$ proves (72). $\square$

4 Compatible intersection, endpoint jet, and restart

Theorem 4.1 (Compatible inverse intersection). *Let $\nu > 0$, $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let (T_*, u, p) be the maximal pressure-complete classical history generated by (u_0, ν) . If $T_* < \infty$, then the family*

$$(\mathcal{R}_{N,M}[u_0; T_*])_{N,M \in \mathbb{N}_0} \quad (74)$$

is the unique datum-generated element of the compatible intersection $\mathcal{I}[u_0]$ from Definition 3.7. It has unique coordinates $(V_n, Q_n)_{n \in \mathbb{N}_0}$ satisfying

$$V_n \in \bigcap_{m \in \mathbb{N}_0} L^2(0, T_*; H_\sigma^m(\mathbb{R}^3)), \quad (75)$$

$$Q_n \in \bigcap_{m \in \mathbb{N}_0} L^1(0, T_*; H^m(\mathbb{R}^3)), \quad (76)$$

and every finite restriction of these coordinates is the corresponding rectangle $\mathcal{R}_{N,M}[u_0; T_]$.*

Proof. Theorem 3.3 places every velocity coordinate of $\mathcal{R}_{N,M}[u_0; T_*]$ in its whole-terminal adjacent spaces. Proposition 3.4 places every retained pressure coordinate in its whole-terminal pressure space. Proposition 3.6 gives

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0; T_*] = \mathcal{R}_{N',M'}[u_0; T_*] \quad (N' \leq N, \ M' \leq M),$$

so (74) belongs to $\mathcal{I}[u_0]$.

Fix n . Every rectangle with $N \geq n$ contains the same distribution $\partial_t^n u$, and every such rectangle contains the same distribution $\partial_t^n p$. Increasing M , using the canonical Sobolev embeddings, gives (75) and (76). Any datum-generated compatible family has these same finite restrictions by Proposition 3.6; equality in every finite restriction gives equality of every distributional coordinate. $\square$

Corollary 4.2 (Whole-terminal pressure-complete mixed jet). *Under the hypotheses of Theorem 4.1, for every $r, m, n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$u \in H^r(0, T_*; H_\sigma^m(\mathbb{R}^3)), \quad (77)$$

$$J_{n,\alpha} := \partial_x^\alpha V_n \in L^2(0, T_*; H^{m+1}(\mathbb{R}^3)), \quad (78)$$

$$J_{n+1,\alpha} \in L^2(0, T_*; H^{m-1}(\mathbb{R}^3)), \quad (79)$$

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n \in L^1(0, T_*; H^m(\mathbb{R}^3)). \quad (80)$$

The following finite bounds hold:

$$\sum_{j=0}^r \|V_j\|_{L^2(0, T_*; H^m)}^2 \leq \mathfrak{R}_{r,m}^*[u_0, \nu, T_*], \quad (81)$$

$$\|J_{n,\alpha}\|_{L^2(0, T_*; H^{m+1})}^2 + \|J_{n+1,\alpha}\|_{L^2(0, T_*; H^{m-1})}^2 \leq \mathfrak{R}_{n,m+|\alpha|}^*[u_0, \nu, T_*], \quad (82)$$

$$\|\Pi_{n,\alpha}\|_{L^1(0, T_*; H^m)} \leq C_{m+|\alpha|}^p (2^{n+1} - 1) \mathfrak{R}_{n,m+|\alpha|}^*[u_0, \nu, T_*]. \quad (83)$$

Every mixed row satisfies the recurrences in Proposition 2.4 on $(0, T_*)$.

Proof. Equation (77) and (81) are Proposition 3.8. The Fourier multiplier bound

$$\|\partial_x^\alpha f\|_{H^s} \leq \|f\|_{H^{s+|\alpha|}}$$

and the rectangle $(n, m + |\alpha|)$ give (78)–(82). Apply the same multiplier bound to the pressure completion in Proposition 3.4 to obtain (80) and (83). Proposition 2.4 supplies the stated identities. $\square$

Theorem 4.3 (Compatible whole-terminal Sobolev traces). *Assume the hypotheses of Theorem 4.1. For $N, m \in \mathbb{N}_0$, define*

$$\mathfrak{E}_{N,m} := (1 + T_*^{-1})^{1/2} (\mathfrak{R}_{N,m}^*[u_0, \nu, T_*])^{1/2}. \quad (84)$$

If $0 \leq n \leq N$ and $\alpha \in \mathbb{N}_0^3$, then

$$V_n \in C([0, T_*]; H_\sigma^m(\mathbb{R}^3)), \quad \sup_{0 \leq t \leq T_*} \|V_n(t)\|_{H^m} \leq \mathfrak{E}_{N,m}, \quad (85)$$

$$J_{n,\alpha} \in C([0, T_*]; H^m(\mathbb{R}^3)), \quad \sup_{0 \leq t \leq T_*} \|J_{n,\alpha}(t)\|_{H^m} \leq \mathfrak{E}_{N,m+|\alpha|}. \quad (86)$$

There are unique values

$$V_n^* := \lim_{t \uparrow T_*} V_n(t) \in H^\infty(\mathbb{R}^3), \quad (87)$$

$$J_{n,\alpha}^* := \lim_{t \uparrow T_*} J_{n,\alpha}(t) = \partial_x^\alpha V_n^* \in H^\infty(\mathbb{R}^3), \quad (88)$$

such that, for $k \geq m$, the image of the H^k trace under $H^k \hookrightarrow H^m$ is the H^m trace. Quantitatively,

$$\|V_n^*\|_{H^m} \leq \mathfrak{E}_{n,m}, \quad (89)$$

$$\|V_n(t) - V_n^*\|_{H^m} \leq (T_* - t) \mathfrak{E}_{n+1,m}, \quad 0 \leq t \leq T_*, \quad (90)$$

$$\|J_{n,\alpha}^*\|_{H^m} \leq \mathfrak{E}_{n,m+|\alpha|}, \quad (91)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}^*\|_{H^m} \leq (T_* - t) \mathfrak{E}_{n+1,m+|\alpha|}. \quad (92)$$

In particular,

$$u_* := V_0^* \in H_\sigma^\infty(\mathbb{R}^3), \quad u \in \bigcap_{a,m \in \mathbb{N}_0} C^a([0, T_*]; H_\sigma^m(\mathbb{R}^3)), \quad \partial_t^a u(T_*) = V_a^*. \quad (93)$$

Proof. For $0 \leq n \leq N$, Theorem 3.3 gives

$$V_n \in L^2(0, T_*; H^{m+1}), \quad \partial_t V_n = V_{n+1} \in L^2(0, T_*; H^{m-1}).$$

Lemma 1.6 gives the first membership in (85) and

$$\begin{aligned} \sup_{0 \leq t \leq T_*} \|V_n(t)\|_{H^m}^2 &\leq T_*^{-1} \|V_n\|_{L^2 H^m}^2 + 2 \|V_n\|_{L^2 H^{m+1}} \|V_{n+1}\|_{L^2 H^{m-1}} \\ &\leq (1 + T_*^{-1}) \mathfrak{R}_{N,m}^*[u_0, \nu, T_*]. \end{aligned}$$

This proves the bound in (85). Apply the same lemma to $J_{n,\alpha}$, using (82), to prove (86).

For $k \geq m$, the H^k -continuous and H^m -continuous representatives equal the same preterminal distribution. After applying $H^k \hookrightarrow H^m$, uniqueness of continuous representatives makes them equal on $[0, T_*]$. Their endpoint values are compatible and define (87)–(88). Spatial differentiation is continuous between the corresponding finite Sobolev spaces, so $J_{n,\alpha}^* = \partial_x^\alpha V_n^*$.

Corollary 4.2 gives $V_n, V_{n+1} \in L^1(0, T_*; H^m)$ and $\partial_t V_n = V_{n+1}$ in distributions. The Banach-valued fundamental identity gives

$$V_n(t) - V_n(s) = \int_s^t V_{n+1}(\tau) d\tau \quad \text{in } H^m, \quad 0 \leq s \leq t \leq T_*. \quad (94)$$

The continuous representative of V_{n+1} is bounded by $\mathfrak{E}_{n+1,m}$. Set $t = T_*$ in (94) to prove (90); spatial differentiation gives (92). The remaining size estimates follow from (85)–(86).

The divergence map $H^{m+1} \rightarrow H^m$ is continuous, so $\nabla \cdot u_* = 0$. For each a, m , the curves $V_0, \dots, V_{a+1}$ are continuous in H^m , and (94) identifies the derivative of V_j with V_{j+1} . Induction in j proves (93). $\square$

Theorem 4.4 (Pressure-complete endpoint jet). *Assume the hypotheses of Theorem 4.1. For $n \in \mathbb{N}_0$, define*

$$Q_n^* := R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^* V_{n-a,j}^*. \quad (95)$$

Then $Q_n^* \in H^\infty(\mathbb{R}^3)$, and, for every $s \in \mathbb{N}_0$,

$$\|Q_n^*\|_{H^s} \leq 9C_s^{\text{alg}} 2^n \mathfrak{E}_{n,s+1}^2, \quad (96)$$

$$\|Q_n(t) - Q_n^*\|_{H^s} \leq 9C_s^{\text{alg}} 2^{n+1} (T_* - t) \mathfrak{E}_{n+1,s+1}^2. \quad (97)$$

For $\alpha \in \mathbb{N}_0^3$,

$$\Pi_{n,\alpha}^* := \partial_x^\alpha Q_n^* = \lim_{t \uparrow T_*} \Pi_{n,\alpha}(t) \quad \text{in } H^m(\mathbb{R}^3) \text{ for every } m \in \mathbb{N}_0. \quad (98)$$

More precisely, for every $m \in \mathbb{N}_0$,

$$\|\Pi_{n,\alpha}^*\|_{H^m} \leq 9C_{m+|\alpha|}^{\text{alg}} 2^n \mathfrak{E}_{n,m+|\alpha|+1}^2, \quad (99)$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}^*\|_{H^m} \leq 9C_{m+|\alpha|}^{\text{alg}} 2^{n+1} (T_* - t) \mathfrak{E}_{n+1,m+|\alpha|+1}^2. \quad (100)$$

The endpoint rows satisfy, for every $n, q \in \mathbb{N}_0$,

$$V_{n+1}^* = \nu \Delta V_n^* - \sum_{a=0}^n \binom{n}{a} (V_a^* \cdot \nabla) V_{n-a}^* - \nabla Q_n^* \quad \text{in } H^q(\mathbb{R}^3). \quad (101)$$

The family $(V_n^*, Q_n^*)_{n \in \mathbb{N}_0}$ is the unique pressure-normalized endpoint jet rooted at u_* . Moreover,

$$p \in \bigcap_{a,m \in \mathbb{N}_0} C^a([0, T_*]; H^m(\mathbb{R}^3)), \quad \partial_t^a p(T_*) = Q_a^*. \quad (102)$$

Proof. Lemmas 1.3 and 1.4, followed by (85), give

$$\begin{aligned}\|Q_n^*\|_{H^s} &\leq 9C_s^{\text{alg}} \sum_{a=0}^n \binom{n}{a} \|V_a^*\|_{H^{s+1}} \|V_{n-a}^*\|_{H^{s+1}} \\ &\leq 9C_s^{\text{alg}} 2^n \mathfrak{E}_{n,s+1}^2.\end{aligned}$$

For each product in $Q_n(t) - Q_n^*$, insert $V_a^* V_{n-a}(t)$, use (90), and bound every retained factor by $\mathfrak{E}_{n+1,s+1}$. The two product differences and $\sum_{a=0}^n \binom{n}{a} = 2^n$ prove (97). The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^m} \leq \|f\|_{H^{m+|\alpha|}}$$

applied to (96) and (97) proves (99) and (100). The latter tends to zero as $t \uparrow T_*$, which proves (98) in every H^m .

Fix n, q . The trace of V_n in H^{q+2} carries the diffusion term in Proposition 2.4 to H^q at T_* . The product estimate at order $q+1$, using velocity convergence in H^{q+2} , carries every transport term to H^q . Equation (97) at order $q+1$ carries the pressure gradient to H^q . The left side converges to V_{n+1}^* in H^q , which proves (101).

The value $V_0^* = u_*$ fixes Q_0^* , and the velocity recurrence fixes V_1^* . Inductively, $V_0^*, \dots, V_n^*$ fix Q_n^* , which fixes V_{n+1}^* . This proves uniqueness. Extend each Q_n to T_* by $Q_n(T_*) := Q_n^*$. On $(0, T_*)$, $Q_n = \partial_t^n p$, so $\partial_t Q_n = Q_{n+1}$ in distributions. Both rows are continuous in every H^m by (97). Hence, for $0 \leq s \leq t \leq T_*$,

$$Q_n(t) - Q_n(s) = \int_s^t Q_{n+1}(\tau) d\tau \quad \text{in } H^m(\mathbb{R}^3). \quad (103)$$

Indeed, the identity holds for $t < T_*$, and the endpoint bounds permit passage to $t = T_*$. Thus $Q_n \in C^1([0, T_*]; H^m)$ with derivative Q_{n+1} . Induction gives (102). $\square$

Lemma 4.5 (Late-slice overlap uniqueness). *Assume the hypotheses of Theorem 4.1. Define*

$$\delta_0 := \min\{1, c_\nu(1 + \mathfrak{E}_{0,3})^{-2}\} > 0, \quad (104)$$

where c_ν is the constant in Theorem 1.9. For every $t_0 \in (\max\{0, T_* - \delta_0/2\}, T_*)$, let (w^{t_0}, q^{t_0}) be the pressure-normalized local solution generated by $u(t_0)$, translated to $[t_0, t_0 + \delta_0]$. Let (w^*, q^*) be the pressure-normalized local solution generated by u_* , translated to $[T_*, T_* + \delta_0]$. Then

$$(w^{t_0}, q^{t_0}) = (u, p) \quad \text{on } [t_0, T_*], \quad (105)$$

$$(w^{t_0}, q^{t_0}) = (w^*, q^*) \quad \text{on } [T_*, t_0 + \delta_0]. \quad (106)$$

Proof. Equation (85) gives

$$\|u(t_0)\|_{H^3} \leq \mathfrak{E}_{0,3}, \quad \|u_*\|_{H^3} \leq \mathfrak{E}_{0,3}.$$

Theorem 1.9, translated in time, gives both local solutions on the stated intervals. Its strong uniqueness gives (105). Continuity in H^3 gives

$$w^{t_0}(T_*) = \lim_{t \uparrow T_*} u(t) = u_*.$$

The two local solutions starting at T_* have the same datum and the same pressure normalization, so strong uniqueness gives (106). The right overlap has positive length because $t_0 + \delta_0 > T_* + \delta_0/2$. $\square$

Theorem 4.6 (Same-history endpoint restart). *Assume the hypotheses of Theorem 4.1. There is a pressure-normalized classical solution $(\tilde{u}, \tilde{p})$ on $[0, T_* + \delta_0/2] \times \mathbb{R}^3$ such that*

$$(\tilde{u}, \tilde{p})|_{[0, T_*] \times \mathbb{R}^3} = (u, p), \quad (107)$$

$$\tilde{u}, \tilde{p} \in C^\infty([0, T_* + \delta_0/2] \times \mathbb{R}^3), \quad (108)$$

$$\partial_t^n \tilde{u}(T_*) = V_n^*, \quad \partial_t^n \tilde{p}(T_*) = Q_n^* \quad (n \in \mathbb{N}_0). \quad (109)$$

The extension is the unique local continuation of the same maximal history.

Proof. Let (w^*, q^*) be the endpoint solution from Lemma 4.5, and set

$$W_n^+ := \partial_t^n w^*(T_*), \quad P_n^+ := \partial_t^n q^*(T_*).$$

Theorem 1.9 and Proposition 2.4 give

$$\begin{aligned} P_n^+ &= R_i R_j \sum_{a=0}^n \binom{n}{a} W_{a,i}^+ W_{n-a,j}^+, \\ W_{n+1}^+ &= \nu \Delta W_n^+ - \sum_{a=0}^n \binom{n}{a} (W_a^+ \cdot \nabla) W_{n-a}^+ - \nabla P_n^+. \end{aligned}$$

Since $W_0^+ = u_* = V_0^*$, induction with (95) and (101) gives

$$W_n^+ = V_n^*, \quad P_n^+ = Q_n^*, \quad n \in \mathbb{N}_0. \quad (110)$$

Define

$$(\tilde{u}, \tilde{p})(t) := \begin{cases} (u, p)(t), & 0 \leq t < T_*, \\ (w^*, q^*)(t), & T_* \leq t \leq T_* + \delta_0/2. \end{cases}$$

Equations (93), (102), and (110) match every one-sided time derivative in every finite Sobolev space. Sobolev embedding gives (108), and the pressure-complete equation holds at T_* by (101). Lemma 4.5 identifies the joined pair near T_* with the single local history (w^{t_0}, q^{t_0}) . This proves (107) and local uniqueness of the continuation. $\square$

Corollary 4.7 (Maximal-time contradiction). *The maximal pressure-complete classical history generated by (u_0, ν) cannot have $T_* < \infty$.*

Proof. If $T_* < \infty$, Theorem 3.3 supplies the settled whole-terminal rectangles. Theorems 4.1–4.6 produce a pressure-normalized classical extension to $[0, T_* + \delta_0/2]$, where $\delta_0 > 0$. This contradicts maximality in Definition 2.2. $\square$

Theorem 4.8 (Global smoothness on $\mathbb{R}^3$). *For every $\nu > 0$ and every $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, there exists a pressure-normalized pair*

$$(u, p) \in C^\infty([0, \infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R}) \quad (111)$$

satisfying

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \nu \Delta u, \quad (112)$$

$$\nabla \cdot u = 0, \quad u(0) = u_0, \quad (113)$$

$$p(t) = R_i R_j (u_i(t) u_j(t)), \quad \lim_{|x| \rightarrow \infty} p(t, x) = 0, \quad (114)$$

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds = \frac{1}{2} \|u_0\|_{L^2}^2, \quad t \geq 0. \quad (115)$$

In particular,

$$\sup_{t \geq 0} \|u(t)\|_{L^2(\mathbb{R}^3)}^2 \leq \|u_0\|_{L^2(\mathbb{R}^3)}^2 < \infty. \quad (116)$$

This pair is unique among pressure-normalized solutions whose velocity belongs to $C([0, T]; H_\sigma^3(\mathbb{R}^3))$ for every finite $T > 0$.

Proof. Let

$$\mathcal{T} := \left\{ \tau > 0 : \begin{array}{l} \text{there exists a pressure-complete classical history} \\ (\tau, u^\tau, p^\tau) \text{ generated by } (u_0, \nu) \end{array} \right\},$$

and put

$$T_* := \sup \mathcal{T}.$$

Theorem 1.9 shows that $\mathcal{T}$ is nonempty. Strong uniqueness identifies any two such solutions on their common interval; their union defines a unique solution on $[0, T_*)$. Since $u_0 \in \mathcal{S}_\sigma$, the persistence conclusion of Theorem 1.9, applied on overlapping local intervals, places every strict restriction in the class of Definition 2.2. An extension beyond T_* would restrict to an interval $[0, \tau]$ with $\tau > T_*$, contrary to the definition of the supremum. Thus (T_*, u, p) is maximal. Corollary 4.7 gives $T_* = \infty$.

Fix $T < \infty$. Persistence in Theorem 1.9, iterated along the unique history, retains every finite spatial Sobolev order on $[0, T]$. Proposition 2.4 retains every finite temporal row. Lemma 1.2 gives (111); arbitrariness of T gives the global class. The pressure belongs to $H^s(\mathbb{R}^3)$ for every s , and the Fourier inversion proof of Lemma 1.2 places its continuous representative in $C_0(\mathbb{R}^3)$, which proves the limit in (114). Any second pressure-normalized solution in the stated uniqueness class agrees with u on each finite interval by the full $C([0, T]; H_\sigma^3)$ uniqueness in Theorem 1.9; pressure normalization then identifies its pressure with p . This proves global uniqueness in the asserted class.

Proposition 3.1 holds on every finite interval. Since T was arbitrary, it gives (115) for every $t \geq 0$; (116) follows by discarding the nonnegative dissipation term. $\square$

5 Quantitative consequences of the compatible rectangles

Definition 5.1 (Producer constants). Let $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, $\nu > 0$, and let (u, p) be the datum-generated pressure-normalized solution. Let $0 < T < \infty$ satisfy either $T < T_*$ or $T = T_* < \infty$, and set $I_T = (0, T)$,

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad n \in \mathbb{N}_0.$$

Define the monotone producer constant

$$\mathfrak{P}_{N,M}[u_0, \nu; T] := 1 + \max_{\substack{0 \leq N' \leq N \\ 0 \leq M' \leq M}} \sup_{0 < S < T} \mathfrak{R}_{N',M'}(u; S), \quad (117)$$

and the trace constant

$$\mathfrak{T}_{N,M}[u_0, \nu; T] := (1 + T^{-1})^{1/2} \mathfrak{P}_{N,M}[u_0, \nu; T]^{1/2}. \quad (118)$$

All occurrences of $\mathfrak{P}_{N,M}$ and $\mathfrak{T}_{N,M}$ below carry the arguments $[u_0, \nu; T]$.

Lemma 5.2 (Producer finiteness). *For every $N, M \in \mathbb{N}_0$,*

$$1 \leq \mathfrak{P}_{N,M}[u_0, \nu; T] < \infty. \quad (119)$$

Proof. If $T = T_* < \infty$, equation (58) proves (119). If $T < T_*$, classical persistence on $[0, T]$ gives

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} < \infty \quad (n, m \in \mathbb{N}_0),$$

and (52) proves (119). □

Lemma 5.3 (Fixed analytic constants). *For $r \in \mathbb{N}_0$, set*

$$A_r := 18C_r^{\text{alg}}.$$

Then

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^{r-1}} \leq A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \quad (120)$$

The same constant satisfies

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\nabla R_i R_j(f_i g_j)\|_{H^{r-1}} \leq A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \quad (121)$$

For $s, k \in \mathbb{N}_0$, $2 \leq b \leq \infty$, and $s > k + \frac{3}{2} - \frac{3}{b}$, define

$$d_k := \#\{\alpha \in \mathbb{N}_0^3 : |\alpha| = k\}, \quad \frac{1}{q_b} := \frac{1}{2} - \frac{1}{b},$$

with $q_2 = \infty$, and

$$S_{s,k,b} := d_k^{1/2} (2\pi)^{-3(1/2-1/b)} \left\| |\xi|^k \langle \xi \rangle^{-s} \right\|_{L_\xi^{q_b}}. \quad (122)$$

Then $S_{s,k,b} < \infty$ and

$$\left\| \nabla^k f \right\|_{L^b} \leq S_{s,k,b} \|f\|_{H^s}. \quad (123)$$

For $m \geq 2$, set $L_m := \max\{1, C_m^{\text{alg}}\}$. Then

$$\|fg\|_{H^m} \leq L_m \|f\|_{H^m} \|g\|_{H^m}, \quad (124)$$

$$\|e^{\nu t \Delta} \mathbb{P}\nabla \cdot F\|_{H^m} \leq L_m (\nu t)^{-1/2} \|F\|_{H^m}, \quad t > 0. \quad (125)$$

Proof. Lemmas 1.3 and 1.4 give

$$\|R_i R_j(f_i g_j)\|_{H^r} \leq \sum_{i,j=1}^3 \|f_i g_j\|_{H^r} \leq 9C_r^{\text{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}},$$

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^{r-1}} \leq \|f \otimes g\|_{H^r} \leq 9C_r^{\text{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}.$$

Moreover,

$$\|\nabla R_i R_j(f_i g_j)\|_{H^{r-1}} \leq \|R_i R_j(f_i g_j)\|_{H^r}.$$

This proves (120) and (121).

Let $b' = b/(b-1)$. Interpolation between Fourier inversion $L_\xi^1 \rightarrow L_x^\infty$, with norm $(2\pi)^{-3/2}$, and Plancherel gives

$$\|\mathcal{F}^{-1}h\|_{L_x^b} \leq (2\pi)^{-3(1/2-1/b)} \|h\|_{L_\xi^{b'}}.$$

For $|\alpha| = k$, Hölder's inequality with $1/b' = 1/2 + 1/q_b$ gives

$$\|\partial^\alpha f\|_{L^b} \leq (2\pi)^{-3(1/2-1/b)} \left\| |\xi|^k \langle \xi \rangle^{-s} \right\|_{L^{q_b}} \|f\|_{H^s}.$$

The weight belongs to L^{q_b} under the displayed strict inequality for s ; for $b = 2$, its L^∞ -norm is finite because $s > k$. Summation over $|\alpha| = k$ proves (123).

The first estimate in (124) is Lemma 1.3. For the heat estimate, the symbol of $\mathbb{P}\nabla$, acting from tensors to vectors, has norm at most $|\xi|$, while

$$\sup_{\rho \geq 0} \rho e^{-\nu t \rho^2} = (2e\nu t)^{-1/2}.$$

Plancherel gives

$$\|e^{\nu t \Delta} \mathbb{P}\nabla \cdot F\|_{H^m} \leq (2e\nu t)^{-1/2} \|F\|_{H^m} \leq L_m(\nu t)^{-1/2} \|F\|_{H^m}.$$

□

Definition 5.4 (Canonical datum majorants). For $N, M \in \mathbb{N}_0$, define

$$\mathfrak{B}_{N,M}[u_0, \nu; T] := \inf \{B \geq 0 : \mathfrak{R}_{N,M}(u; S) \leq B \text{ for every } 0 < S < T\}. \quad (126)$$

Monotone convergence gives

$$\mathfrak{B}_{N,M}[u_0, \nu; T] = \begin{cases} \mathfrak{R}_{N,M}(u; T), & T < T_*, \\ \mathfrak{R}_{N,M}^*[u_0, \nu, T_*], & T = T_* < \infty. \end{cases} \quad (127)$$

Consequently,

$$\mathfrak{B}_{N,M}[u_0, \nu; T] = 1 + \max_{\substack{0 \leq N' \leq N \\ 0 \leq M' \leq M}} \mathfrak{B}_{N',M'}[u_0, \nu; T]. \quad (128)$$

For $n, m \in \mathbb{N}_0$, define the initial-jet majorants $\mathfrak{D}_{n,m} = \mathfrak{D}_{n,m}[u_0, \nu]$ by

$$\mathfrak{D}_{0,m} := \|u_0\|_{H^m}, \quad (129)$$

$$\mathfrak{D}_{n+1,m} := \nu \mathfrak{D}_{n,m+2} + A_{m+1} \sum_{a=0}^n \binom{n}{a} \mathfrak{D}_{a,m+2} \mathfrak{D}_{n-a,m+2}. \quad (130)$$

Define the pointwise recursion $\mathfrak{Y}_{n,m} = \mathfrak{Y}_{n,m}[u_0, \nu; T]$ by

$$\mathfrak{Y}_{0,0} := \|u_0\|_2, \quad (131)$$

$$\mathfrak{Y}_{0,m} := \min \left\{ (1 + T^{-1})^{1/2} \mathfrak{B}_{0,m}^{1/2}, \|u_0\|_{H^m} + T^{1/2} \mathfrak{B}_{0,m+1}^{1/2} \right\}, \quad m \geq 1, \quad (132)$$

$$\mathfrak{Y}_{n+1,m} := \nu \mathfrak{Y}_{n,m+2} + A_{m+1} \sum_{a=0}^n \binom{n}{a} \mathfrak{Y}_{a,m+2} \mathfrak{Y}_{n-a,m+2}. \quad (133)$$

For $m \in \{-1\} \cup \mathbb{N}_0$, put

$$m_+ := \max\{m, 0\}, \quad \mu(m) := \max\{m - 1, 0\},$$

and define

$$\tilde{\mathfrak{X}}_{n,m} := \min \left\{ T^{1/2} \mathfrak{Y}_{n,m_+}, \mathfrak{B}_{n,\mu(m)}^{1/2} \right\}. \quad (134)$$

Set

$$\mathfrak{X}_{0,0} := \min \left\{ \tilde{\mathfrak{X}}_{0,0}, T^{1/2} \|u_0\|_2 \right\}, \quad (135)$$

$$\mathfrak{X}_{0,1} := \min \left\{ \tilde{\mathfrak{X}}_{0,1}, \left(T + \frac{1}{2\nu} \right)^{1/2} \|u_0\|_2 \right\}, \quad (136)$$

$$\mathfrak{X}_{n,m} := \tilde{\mathfrak{X}}_{n,m} \quad \text{for all other } (n, m). \quad (137)$$

For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{Z}_{n,m}[u_0, \nu; T] := \min \left\{ \mathfrak{Y}_{n,m}, (1 + T^{-1})^{1/2} \mathfrak{B}_{n,m}^{1/2}, \mathfrak{D}_{n,m} + T^{1/2} \mathfrak{X}_{n+1,m} \right\}. \quad (138)$$

All $\mathfrak{B}, \mathfrak{D}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z}$ in this section carry the arguments $[u_0, \nu; T]$, except that $\mathfrak{D}$ is independent of T .

Lemma 5.5 (Closed velocity majorants). *For every $n, m \in \mathbb{N}_0$,*

$$\|V_n(0)\|_{H^m} \leq \mathfrak{D}_{n,m}, \quad (139)$$

$$\|V_n\|_{L^2(I_T; H^m)} \leq \mathfrak{X}_{n,m}, \quad (140)$$

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq \mathfrak{Z}_{n,m}. \quad (141)$$

For $m = -1$, equation (140) remains valid. Every right-hand side is a finite function of (u_0, ν, n, m, T) .

Proof. Applying $\mathbb{P}$ to the temporal recurrence gives

$$V_{n+1} = \nu \Delta V_n - \mathbb{P} \nabla \cdot \sum_{a=0}^n \binom{n}{a} (V_a \otimes V_{n-a}). \quad (142)$$

Equation (120) at order $m+1$, evaluated at $t=0$, proves (139) by induction through (130). The same estimate, taken pointwise on $[0, T]$, proves by induction that

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq \mathfrak{Y}_{n,m}. \quad (143)$$

The base case $m=0$ is (50). For $m \geq 1$, both terms in (132) are valid: the first is the adjacent trace bound, and the second follows from

$$u(t) = u_0 + \int_0^t V_1(s) \, ds \quad \text{in } H^m.$$

The rectangle with middle order $\mu(m)$, Sobolev nesting, and (143) give respectively

$$\|V_n\|_{L^2(I_T; H^m)} \leq \mathfrak{B}_{n, \mu(m)}^{1/2}, \quad \|V_n\|_{L^2(I_T; H^m)} \leq T^{1/2} \mathfrak{Y}_{n, m+}.$$

This proves (140) through (134); the two exceptional improvements are (50) and (51). The adjacent trace lemma gives

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq (1 + T^{-1})^{1/2} \mathfrak{B}_{n,m}^{1/2},$$

while the Banach-valued fundamental identity and (139) give

$$\sup_{0 \leq t \leq T} \|V_n(t)\|_{H^m} \leq \mathfrak{D}_{n,m} + T^{1/2} \mathfrak{X}_{n+1,m}.$$

Taking the minimum of these bounds and (143) proves (141). $\square$

Definition 5.6 (Closed pressure majorants). For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{Q}_{n,m}^{(1)} := A_m \sum_{a=0}^n \binom{n}{a} \mathfrak{X}_{a,m+1} \mathfrak{X}_{n-a,m+1}, \quad (144)$$

$$\mathfrak{Q}_{n,m}^{(2)} := A_m \sum_{a=0}^n \binom{n}{a} \min \{ \mathfrak{Z}_{a,m+1} \mathfrak{X}_{n-a,m+1}, \mathfrak{X}_{a,m+1} \mathfrak{Z}_{n-a,m+1} \}, \quad (145)$$

$$\mathfrak{Q}_{n,m}^{(\infty)} := A_m \sum_{a=0}^n \binom{n}{a} \mathfrak{Z}_{a,m+1} \mathfrak{Z}_{n-a,m+1}, \quad (146)$$

$$\mathfrak{Q}_{n,m}^{(\text{mod})} := A_m \sum_{a=0}^n \binom{n}{a} (\mathfrak{X}_{a+1,m+1} \mathfrak{Z}_{n-a,m+1} + \mathfrak{Z}_{a,m+1} \mathfrak{X}_{n-a+1,m+1}). \quad (147)$$

These quantities carry the arguments $[u_0, \nu; T]$ through $\mathfrak{X}$ and $\mathfrak{Z}$.

Theorem 5.7 (Quantitative mixed-jet rows and traces). *Let $N, s \in \mathbb{N}_0$, let $0 \leq n \leq N$, and let $\alpha \in \mathbb{N}_0^3$. Put*

$$J_{n,\alpha} := \partial_x^\alpha V_n.$$

Then

$$\|J_{n,\alpha}\|_{L^2(I_T; H^{s+1})}^2 + \|J_{n+1,\alpha}\|_{L^2(I_T; H^{s-1})}^2 \leq \mathfrak{P}_{N,s+|\alpha|}, \quad (148)$$

$$\sup_{0 \leq t \leq T} \|J_{n,\alpha}(t)\|_{H^s} \leq \mathfrak{T}_{N,s+|\alpha|}. \quad (149)$$

There is a unique trace $J_{n,\alpha}^T \in H^s$ at $t = T$, and for $0 \leq r \leq t \leq T$,

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-1}} \leq (t-r)^{1/2} \mathfrak{P}_{N,s+|\alpha|}^{1/2}, \quad (150)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^s} \leq (t-r)^{1/2} \mathfrak{P}_{N,s+|\alpha|+1}^{1/2}, \quad (151)$$

$$\|J_{n,\alpha}^T\|_{H^s} \leq \mathfrak{T}_{N,s+|\alpha|}. \quad (152)$$

If $0 \leq \theta \leq 1$, then

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-\theta}} \leq 2^{1-\theta} \mathfrak{T}_{N,s+|\alpha|}^{1-\theta} \mathfrak{P}_{N,s+|\alpha|}^{\theta/2} (t-r)^{\theta/2}. \quad (153)$$

For $1 \leq a \leq \infty$, $2 \leq b \leq \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$,

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} \mathfrak{T}_{N,q+|\alpha|}, \quad (154)$$

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^2(I_T; L^b)} \leq S_{q,k,b} \mathfrak{P}_{N,q+|\alpha|}^{1/2}, \quad (155)$$

where $T^{1/\infty} = 1$.

Proof. Take the rectangle with middle order $M = s + |\alpha|$. The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^r} \leq \|f\|_{H^{r+|\alpha|}}$$

and (117) give (148). The adjacent-row trace estimate gives

$$\sup_{0 \leq t \leq T} \|J_{n,\alpha}(t)\|_{H^s}^2 \leq T^{-1} \|J_{n,\alpha}\|_{L^2(I_T; H^s)}^2$$

$$\begin{aligned}
& + 2 \|J_{n,\alpha}\|_{L^2(I_T; H^{s+1})} \|J_{n+1,\alpha}\|_{L^2(I_T; H^{s-1})} \\
& \leq (T^{-1} + 1) \mathfrak{P}_{N,s+|\alpha|},
\end{aligned}$$

which proves (149) and (152). The Bochner identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r) = \int_r^t J_{n+1,\alpha}(\tau) d\tau$$

in H^{s-1} , followed by Cauchy–Schwarz, proves (150). Applying the same argument to the rectangle with middle order $s + |\alpha| + 1$ proves (151). Interpolation between H^{s-1} and H^s , together with $\|J(t) - J(r)\|_{H^s} \leq 2\mathfrak{T}_{N,s+|\alpha|}$, proves (153). Sobolev embedding and (149) prove (154); Sobolev embedding and (148) prove (155). $\square$

Theorem 5.8 (Quantitative pressure rows). *Let $N, s \in \mathbb{N}_0$, $0 \leq n \leq N$, and $\alpha \in \mathbb{N}_0^3$. Set*

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n, \quad r := s + |\alpha|.$$

Then

$$\|\Pi_{n,\alpha}\|_{L^1(I_T; H^s)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I_T; H^{s-1})} \leq 2A_r 2^n \mathfrak{P}_{N,r}, \quad (156)$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T; H^s)} \leq A_r 2^n \mathfrak{T}_{N,r+1} \mathfrak{P}_{N,r}^{1/2}, \quad (157)$$

$$\sup_{0 \leq t \leq T} \|\Pi_{n,\alpha}(t)\|_{H^s} \leq A_r 2^n \mathfrak{T}_{N,r+1}^2. \quad (158)$$

There is a unique endpoint pressure row $\Pi_{n,\alpha}^T \in H^s$, and

$$\|\Pi_{n,\alpha}^T\|_{H^s} \leq A_r 2^n \mathfrak{T}_{N,r+1}^2, \quad (159)$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \leq 2^{n+1} A_r \mathfrak{T}_{N,r+1} \mathfrak{P}_{N,r+2}^{1/2} |t - r_0|^{1/2} \quad (160)$$

for $0 \leq r_0 \leq t \leq T$. If $1 \leq a \leq \infty$, $2 \leq b \leq \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$, then

$$\left\| \nabla^k \Pi_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} A_{q+|\alpha|} 2^n \mathfrak{T}_{N,q+|\alpha|+1}^2. \quad (161)$$

Proof. The normalized pressure recurrence is

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}. \quad (162)$$

Equation (120), Cauchy–Schwarz in time, and $\sum_{a=0}^n \binom{n}{a} = 2^n$ give

$$\|Q_n\|_{L^1(I_T; H^r)} \leq A_r 2^n \mathfrak{P}_{N,r}. \quad (163)$$

The gradient estimate in (156) follows from $\|\nabla f\|_{H^{r-1}} \leq \|f\|_{H^r}$. Using one factor in $L_t^\infty H_x^{r+1}$ and the other in $L_t^2 H_x^{r+1}$ proves (157). Using both factors in $L_t^\infty H_x^{r+1}$ proves (158) and (159).

Subtract (162) at times t and r_0 , split each product difference into two terms, apply (151) in H^{r+1} , and use (149). The two product differences and the binomial sum give (160). Equation (161) follows from (158) and Sobolev embedding. $\square$

Proposition 5.9 (Velocity and pressure frequency tails). *Let $P_{\geq K}$ be the Fourier multiplier of $\mathbf{1}_{\{|\xi| \geq K\}}$, where $K \geq 1$. Let $r_1 \in \mathbb{N}$, $s \in \mathbb{N}_0$, $0 \leq n \leq N$, and $\alpha \in \mathbb{N}_0^3$. Then*

$$\|P_{\geq K} J_{n,\alpha}\|_{L^2(I_T; H^s)} \leq K^{-r_1} \mathfrak{P}_{N,s+r_1+|\alpha|}^{1/2}, \quad (164)$$

$$\sup_{0 \leq t \leq T} \|P_{\geq K} J_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} \mathfrak{T}_{N,s+r_1+|\alpha|}, \quad (165)$$

$$\|P_{\geq K} \Pi_{n,\alpha}\|_{L^1(I_T; H^s)} \leq K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{P}_{N,s+r_1+|\alpha|}, \quad (166)$$

$$\sup_{0 \leq t \leq T} \|P_{\geq K} \Pi_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{T}_{N,s+r_1+|\alpha|+1}^2. \quad (167)$$

The pointwise estimates (165) and (167) hold at $t = T$.

Proof. For every $f \in H^{s+r_1}$,

$$\|P_{\geq K} f\|_{H^s} \leq K^{-r_1} \|f\|_{H^{s+r_1}}.$$

Apply this inequality to (148) and (149) at spatial order $s + r_1$ to obtain the two velocity bounds. Since Fourier multipliers commute,

$$\|P_{\geq K} \Pi_{n,\alpha}\|_{H^s} \leq K^{-r_1} \|Q_n\|_{H^{s+r_1+|\alpha|}}.$$

Integration in time followed by (163) at order $s + r_1 + |\alpha|$ gives (166). Applying the tail inequality directly to $\Pi_{n,\alpha}$, followed by (158) with s replaced by $s + r_1$, gives (167). $\square$

Proposition 5.10 (Critical-height bounds). *Define*

$$E_\sigma(t) := \frac{1}{2} \|\Lambda^\sigma u(t)\|_2^2, \quad H(t) := E_{1/2}(t),$$

and

$$E_{j+1/2, \geq K}(t) := \frac{1}{2} \int_{|\xi| \geq K} |\xi|^{2j+1} |\widehat{u}(t, \xi)|^2 d\xi.$$

For $j \in \mathbb{N}_0$,

$$\sup_{0 \leq t \leq T} E_{j+1/2}(t) \leq \frac{1}{2} \mathfrak{T}_{0,j+1}^2, \quad (168)$$

$$\int_0^T E_{j+1/2}(t) dt \leq \frac{1}{2} \mathfrak{P}_{0,j}. \quad (169)$$

If $m > j + \frac{1}{2}$ is an integer and $K \geq 1$, then

$$\sup_{0 \leq t \leq T} E_{j+1/2, \geq K}(t) \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{T}_{0,m}^2, \quad (170)$$

$$\int_0^T E_{j+1/2, \geq K}(t) dt \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{P}_{0,m-1}. \quad (171)$$

For $k \in \mathbb{N}$,

$$\|H^{(k)}\|_{L^1(0,T)} \leq 2^{k-1} \mathfrak{P}_{k,0}. \quad (172)$$

In particular,

$$\|H'\|_{L^1(0,T)} \leq \frac{1}{2} \mathfrak{P}_{0,2}, \quad (173)$$

$$\int_0^T \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \mathfrak{P}_{0,1}, \quad (174)$$

$$\left\| \mathcal{P}_{1/2} \right\|_{L^1(0,T)} \leq \frac{1}{2} \mathfrak{P}_{0,2} + \nu \mathfrak{P}_{0,1}, \quad (175)$$

$$|H(T) - H(t)| \leq \mathfrak{T}_{0,1} \mathfrak{P}_{0,2}^{1/2} (T-t)^{1/2}, \quad 0 \leq t \leq T, \quad (176)$$

where $\mathcal{P}_{1/2} = H' + \nu \left\| \Lambda^{3/2} u \right\|_2^2$.

Proof. Equations (168) and (169) follow from $\dot{H}^{j+1/2} \leftrightarrow H^{j+1}$, (149), and (148). On $|\xi| \geq K$,

$$|\xi|^{2j+1} \leq K^{2j+1-2m} |\xi|^{2m},$$

which proves (170) and (171).

The identity

$$H^{(k)}(t) = \frac{1}{2} \sum_{a=0}^k \binom{k}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{k-a}(t) \right\rangle$$

and Cauchy–Schwarz in time give (172). At $k = 1$, the two equal terms give

$$\left\| H' \right\|_{L^1(0,T)} \leq \|u_t\|_{L^2(I_T; \dot{H}^{1/2})} \|u\|_{L^2(I_T; \dot{H}^{1/2})} \leq \frac{1}{2} \mathfrak{P}_{0,2}.$$

Equation (174) follows from the upper velocity row of the $(0, 1)$ rectangle. The identity defining $\mathcal{P}_{1/2}$ gives (175). Finally,

$$\|u(T) - u(t)\|_{\dot{H}^{1/2}} \leq \|u(T) - u(t)\|_{H^1} \leq (T-t)^{1/2} \|u_t\|_{L^2(t,T; H^1)} \leq (T-t)^{1/2} \mathfrak{P}_{0,2}^{1/2},$$

and

$$|H(T) - H(t)| \leq \frac{1}{2} (\|u(T)\|_{\dot{H}^{1/2}} + \|u(t)\|_{\dot{H}^{1/2}}) \|u(T) - u(t)\|_{\dot{H}^{1/2}}.$$

Equation (149) bounds each retained velocity factor, so these two estimates prove (176). $\square$

Proposition 5.11 (Continuation norms). *The finite-horizon solution satisfies*

$$\|u\|_{L^2(I_T; L^\infty)} \leq S_{2,0,\infty} \mathfrak{P}_{0,1}^{1/2}, \quad (177)$$

$$\int_0^T \|\nabla u(t)\|_{L^\infty} dt \leq S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}^{1/2}, \quad (178)$$

$$\int_0^T \|\nabla \times u(t)\|_{L^\infty} dt \leq 2S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}^{1/2}. \quad (179)$$

For every integer $m \geq 3$, set $K_m := C_m^{\text{com}}$. Then

$$|\operatorname{Re} \langle A^m \mathbb{P} \nabla \cdot (u \otimes u), A^m u \rangle_{L^2}| \leq K_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2. \quad (180)$$

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^m}^2 \leq \|u_0\|_{H^m}^2 \exp \left(2K_m S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}^{1/2} \right). \quad (181)$$

Proof. Sobolev embedding and the upper rows of the $(0, 1)$ and $(0, 2)$ rectangles give (177) and (178); the pointwise bound $\|\nabla \times u\|_\infty \leq 2 \|\nabla u\|_\infty$ gives (179). The H^m energy identity and (180) give

$$\frac{1}{2} \frac{d}{dt} \|u\|_{H^m}^2 + \nu \|\nabla u\|_{H^m}^2 \leq K_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2,$$

and hence

$$\frac{d}{dt} \|u\|_{H^m}^2 \leq 2K_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2.$$

Gronwall's inequality and (178) prove (181). $\square$

Theorem 5.12 (Quantitative endpoint restart). *Assume $T = T_* < \infty$, and set*

$$u_T := u_* := V_0^*, \quad V_n^T := V_n^*, \quad Q_n^T := Q_n^*.$$

Define

$$\begin{aligned} \mathcal{L}_T &:= \left\{ \delta > 0 : \begin{array}{l} \exists (z, \pi) \text{ pressure-normalized on } [T, T + \delta], \\ z(T) = u_T, \quad z \in C([T, T + \delta]; H_\sigma^3) \end{array} \right\}, \\ \delta_T^{\max} &:= \sup \mathcal{L}_T. \end{aligned} \quad (182)$$

Define

$$\delta_{\text{qc}}[u_0, \nu; T] := \min \left\{ 1, c_\nu (1 + \mathfrak{Z}_{0,3}[u_0, \nu; T])^{-2} \right\}. \quad (183)$$

The pressure-normalized solution (w, q) with $w(T) = u_T$ exists uniquely on $[T, T + \delta_{\text{qc}}]$, and, for every integer $m \geq 3$,

$$w \in C([T, T + \delta_{\text{qc}}]; H_\sigma^m) \cap L^2(T, T + \delta_{\text{qc}}; H^{m+1}), \quad w_t \in L^2(T, T + \delta_{\text{qc}}; H^{m-1}). \quad (184)$$

At the base restart order,

$$\begin{aligned} &\|w\|_{C([T, T + \delta_{\text{qc}}]; H^3)} + \|w\|_{L^2(T, T + \delta_{\text{qc}}; H^4)} + \|w_t\|_{L^2(T, T + \delta_{\text{qc}}; H^2)} \\ &\leq 2L_\nu \mathfrak{Z}_{0,3}[u_0, \nu; T]. \end{aligned} \quad (185)$$

Set

$$\mathfrak{A}_T := 2S_{3,1,\infty} L_\nu \delta_{\text{qc}} \mathfrak{Z}_{0,3}[u_0, \nu; T]. \quad (186)$$

For every integer $m \geq 3$,

$$\sup_{T \leq t \leq T + \delta_{\text{qc}}} \|w(t)\|_{H^m}^2 \leq \mathfrak{Z}_{0,m}[u_0, \nu; T]^2 \exp(2K_m \mathfrak{A}_T). \quad (187)$$

Moreover, $w, q \in C^\infty([T, T + \delta_{\text{qc}}] \times \mathbb{R}^3)$. Define

$$W_n^T := \partial_t^n w(T), \quad P_n^T := \partial_t^n q(T).$$

Then

$$W_n^T = V_n^T, \quad P_n^T = Q_n^T, \quad n \in \mathbb{N}_0. \quad (188)$$

The maximal restart lifespan $\delta_T^{\max}$ obeys

$$\delta_T^{\max} \geq \delta_{\text{qc}}[u_0, \nu; T] > 0. \quad (189)$$

Proof. Equations (89) and (141) give

$$\|u_T\|_{H^3} \leq \mathfrak{Z}_{0,3}.$$

Consequently,

$$\delta_{\text{qc}}[u_0, \nu; T] \leq \min\{1, c_\nu (1 + \|u_T\|_{H^3})^{-2}\}.$$

Theorem 1.9, translated to time T , gives a solution on the interval in the statement. The fixed-point ball in its proof gives (185). Hence

$$\int_T^{T + \delta_{\text{qc}}} \|\nabla w(t)\|_\infty dt \leq S_{3,1,\infty} \delta_{\text{qc}} \|w\|_{C([T, T + \delta_{\text{qc}}]; H^3)} \leq \mathfrak{A}_T.$$

The H^m energy estimate in Proposition 5.11, now on the restart interval, and $\|u_T\|_{H^m} \leq \mathfrak{Z}_{0,m}$ prove (187). Equation (93) gives $u_T \in H_\sigma^\infty$. The persistence assertion in Theorem 1.9 proves (184) for

every $m \geq 3$ and gives the stated smoothness before the endpoint rows are defined. The full-class uniqueness proved in Theorem 1.9 gives the uniqueness asserted here.

The endpoint jet from Theorem 4.4 satisfies

$$Q_n^T = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^T V_{n-a,j}^T,$$

$$V_{n+1}^T = \nu \Delta V_n^T - \sum_{a=0}^n \binom{n}{a} (V_a^T \cdot \nabla) V_{n-a}^T - \nabla Q_n^T.$$

The smooth right-hand solution satisfies the same two recurrences with (W_n^T, P_n^T) . Since $W_0^T = u_T = V_0^T$, induction in n proves (188). Theorem 4.6 and strong uniqueness identify this solution with the continuation of the same preterminal history. Its existence on the displayed interval proves (189). $\square$

Theorem 5.13 (Sharp datum-indexed finite-horizon estimates). *Let $n, s, k, q \in \mathbb{N}_0$, $\alpha \in \mathbb{N}_0^3$, and set $r := s + |\alpha|$. Then*

$$\|J_{n,\alpha}\|_{L^2(I_T; H^{s+1})} \leq \mathfrak{X}_{n,r+1}, \quad (190)$$

$$\|J_{n+1,\alpha}\|_{L^2(I_T; H^{s-1})} \leq \mathfrak{X}_{n+1,r-1}, \quad (191)$$

$$\sup_{0 \leq t \leq T} \|J_{n,\alpha}(t)\|_{H^s} \leq \mathfrak{Z}_{n,r}, \quad (192)$$

$$\|J_{n,\alpha}^T\|_{H^s} \leq \mathfrak{Z}_{n,r}, \quad (193)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-1}} \leq |t - r_0|^{1/2} \mathfrak{X}_{n+1,r-1}, \quad (194)$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^s} \leq |t - r_0|^{1/2} \mathfrak{X}_{n+1,r}, \quad (195)$$

for $0 \leq r_0 \leq t \leq T$. For $0 \leq \theta \leq 1$,

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-\theta}} \leq 2^{1-\theta} \mathfrak{Z}_{n,r}^{1-\theta} \mathfrak{X}_{n+1,r-1}^\theta |t - r_0|^{\theta/2}. \quad (196)$$

If $1 \leq a \leq \infty$, $2 \leq b \leq \infty$, and $q > k + \frac{3}{2} - \frac{3}{b}$, then

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} \mathfrak{Z}_{n,q+|\alpha|}, \quad (197)$$

$$\left\| \nabla^k J_{n,\alpha} \right\|_{L^2(I_T; L^b)} \leq S_{q,k,b} \mathfrak{X}_{n,q+|\alpha|}, \quad (198)$$

where $T^{1/\infty} = 1$.

The pressure rows obey

$$\|\Pi_{n,\alpha}\|_{L^1(I_T; H^s)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I_T; H^{s-1})} \leq \mathfrak{Q}_{n,r}^{(1)}, \quad (199)$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T; H^s)} \leq \mathfrak{Q}_{n,r}^{(2)}, \quad (200)$$

$$\sup_{0 \leq t \leq T} \|\Pi_{n,\alpha}(t)\|_{H^s} \leq \mathfrak{Q}_{n,r}^{(\infty)}, \quad (201)$$

$$\|\Pi_{n,\alpha}^T\|_{H^s} \leq \mathfrak{Q}_{n,r}^{(\infty)}, \quad (202)$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \leq |t - r_0|^{1/2} \mathfrak{Q}_{n,r}^{(\text{mod})}. \quad (203)$$

Under the preceding restrictions on a, b, k, q ,

$$\left\| \nabla^k \Pi_{n,\alpha} \right\|_{L^a(I_T; L^b)} \leq S_{q,k,b} T^{1/a} \mathfrak{Q}_{n,q+|\alpha|}^{(\infty)}. \quad (204)$$

Let $r_1 \in \mathbb{N}$ and $K \geq 1$. Then

$$\|P_{\geq K} J_{n,\alpha}\|_{L^2(I_T; H^s)} \leq K^{-r_1} \mathfrak{X}_{n,s+r_1+|\alpha|}, \quad (205)$$

$$\sup_{0 \leq t \leq T} \|P_{\geq K} J_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} \mathfrak{Z}_{n,s+r_1+|\alpha|}, \quad (206)$$

$$\|P_{\geq K} \Pi_{n,\alpha}\|_{L^1(I_T; H^s)} \leq K^{-r_1} \mathfrak{Q}_{n,s+r_1+|\alpha|}^{(1)}, \quad (207)$$

$$\sup_{0 \leq t \leq T} \|P_{\geq K} \Pi_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} \mathfrak{Q}_{n,s+r_1+|\alpha|}^{(\infty)}. \quad (208)$$

For $j \in \mathbb{N}_0$,

$$\sup_{0 \leq t \leq T} E_{j+1/2}(t) \leq \frac{1}{2} \mathfrak{Z}_{0,j+1}^2, \quad (209)$$

$$\int_0^T E_{j+1/2}(t) dt \leq \frac{1}{2} \mathfrak{X}_{0,j+1}^2. \quad (210)$$

If $m > j + \frac{1}{2}$ is an integer, then

$$\sup_{0 \leq t \leq T} E_{j+1/2, \geq K}(t) \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{Z}_{0,m}^2, \quad (211)$$

$$\int_0^T E_{j+1/2, \geq K}(t) dt \leq \frac{1}{2} K^{2j+1-2m} \mathfrak{X}_{0,m}^2. \quad (212)$$

For $k \in \mathbb{N}$,

$$\|H^{(k)}\|_{L^1(0,T)} \leq \frac{1}{2} \sum_{a=0}^k \binom{k}{a} \mathfrak{X}_{a,1} \mathfrak{X}_{k-a,1}. \quad (213)$$

In particular,

$$\|H'\|_{L^1(0,T)} \leq \mathfrak{X}_{1,1} \mathfrak{X}_{0,1}, \quad (214)$$

$$\int_0^T \left\| \Lambda^{3/2} u(t) \right\|_2^2 dt \leq \mathfrak{X}_{0,2}^2, \quad (215)$$

$$\|\mathcal{P}_{1/2}\|_{L^1(0,T)} \leq \mathfrak{X}_{1,1} \mathfrak{X}_{0,1} + \nu \mathfrak{X}_{0,2}^2, \quad (216)$$

$$|H(T) - H(t)| \leq \mathfrak{Z}_{0,1} \mathfrak{X}_{1,1} (T-t)^{1/2}, \quad 0 \leq t \leq T. \quad (217)$$

Finally,

$$\|u\|_{L^2(I_T; L^\infty)} \leq S_{2,0,\infty} \mathfrak{X}_{0,2}, \quad (218)$$

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq S_{3,1,\infty} T^{1/2} \mathfrak{X}_{0,3}, \quad (219)$$

$$\int_0^T \|\nabla \times u(t)\|_\infty dt \leq 2S_{3,1,\infty} T^{1/2} \mathfrak{X}_{0,3}, \quad (220)$$

$$\sup_{0 \leq t \leq T} \|u(t)\|_{H^m}^2 \leq \|u_0\|_{H^m}^2 \exp\left(2K_m S_{3,1,\infty} T^{1/2} \mathfrak{X}_{0,3}\right), \quad m \geq 3. \quad (221)$$

Every right-hand side is a finite function of the displayed datum, viscosity, finite derivative indices, frequency threshold, and time horizon.

Proof. The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^\ell} \leq \|f\|_{H^{\ell+|\alpha|}} ,$$

followed by Lemma 5.5, proves (190)–(193). The Banach identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r_0) = \int_{r_0}^t J_{n+1,\alpha}(\tau) \, d\tau$$

in the two indicated Sobolev spaces and Cauchy–Schwarz prove (194) and (195). Interpolation between the low modulus and

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^s} \leq 2\mathfrak{Z}_{n,r}$$

proves (196). The fixed Sobolev embedding constant in (123) proves (197) and (198).

Apply the recurrence

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}$$

term by term at order r , using (121). Cauchy–Schwarz in time gives (199); placing either factor in $L_t^\infty H_x^{r+1}$ gives (200); placing both factors there gives (201) and (202). After subtracting the recurrence at times t and r_0 , split each product difference into two terms and apply (195); this gives (203). Sobolev embedding proves (204).

For every $f \in H^{\ell+r_1}$,

$$\|P_{\geq K} f\|_{H^\ell} \leq K^{-r_1} \|f\|_{H^{\ell+r_1}} .$$

Applying this inequality to the preceding velocity and pressure estimates proves (205)–(208).

The inequalities

$$\|f\|_{\dot{H}^{j+1/2}} \leq \|f\|_{H^{j+1}} , \quad |\xi|^{2j+1} \leq K^{2j+1-2m} |\xi|^{2m} \quad (|\xi| \geq K),$$

give (209)–(212). The exact identity

$$H^{(k)}(t) = \frac{1}{2} \sum_{a=0}^k \binom{k}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{k-a}(t) \right\rangle_{L^2}$$

and Cauchy–Schwarz in time give (213) and (214). The next two estimates follow from

$$\mathcal{P}_{1/2} = H' + \nu \left\| \Lambda^{3/2} u \right\|_2^2 .$$

Finally,

$$|H(T) - H(t)| \leq \frac{1}{2} (\|u(T)\|_{\dot{H}^{1/2}} + \|u(t)\|_{\dot{H}^{1/2}}) \|u(T) - u(t)\|_{\dot{H}^{1/2}} ,$$

and the Banach identity in H^1 prove (217).

Sobolev embedding and (140) prove (218)–(220). Lemma 1.8 gives

$$\frac{d}{dt} \|u\|_{H^m}^2 \leq 2K_m \|\nabla u\|_\infty \|u\|_{H^m}^2 .$$

Gronwall’s inequality and (219) prove (221). □